

第六章. 不确定关系与守恒定律

一. 角动量关系.

(1) 定义: $\hat{L} = \hat{r} \times \hat{p} = -i\hbar \vec{r} \times \vec{\nabla}$

球坐标系中: $\hat{L}_z = -i\hbar \frac{\partial}{\partial \varphi}$

(2) 对易关系:

①. $[\hat{r}_\alpha, \hat{L}_\beta] = i\hbar \epsilon_{\alpha\beta\gamma} \hat{r}_\gamma$

$\epsilon = \begin{cases} 1 & \text{1 2 3} \\ -1 & \text{2 1 3} \\ 0 & \text{其它} \end{cases} \rightarrow \alpha \neq \beta$

②. $[\hat{p}_\alpha, \hat{L}_\beta] = i\hbar \epsilon_{\alpha\beta\gamma} \hat{p}_\gamma$

③. $[\hat{L}_\alpha, \hat{L}_\beta] = i\hbar \epsilon_{\alpha\beta\gamma} \hat{L}_\gamma$

$\hat{L} \times \hat{L} = i\hbar \hat{L}$ (与经典不同)

④. $[\hat{L}_\alpha, \hat{r}^2] = 0$

$[\hat{L}_\alpha, \hat{r}^2] = 0$

$[\hat{L}_\alpha, \hat{p}^2] = 0$

$[\hat{L}_\alpha, \hat{r} \cdot \hat{p}] = 0$

(3) $\hat{L}_z$ 的本征值方程:

$\hat{L}_z |\psi\rangle = \lambda |\psi\rangle$

在坐标表象下: $\langle r | \psi \rangle = \psi(r)$

$\therefore \hat{L}_z \psi = \lambda \psi$

又 $\hat{L}_z = -i\hbar \frac{\partial}{\partial \varphi}$

$\Rightarrow \psi = A e^{\frac{i}{\hbar} \lambda \varphi}$

下面求 λ 和 A : 1. 由周期性边界条件: $\psi(\varphi) = \psi(\varphi + 2\pi)$

$\therefore A e^{\frac{i}{\hbar} \lambda \varphi} = A e^{\frac{i}{\hbar} \lambda (\varphi + 2\pi)}$

$\therefore \frac{\lambda 2\pi}{\hbar} = 2\pi m$

$\Rightarrow \lambda = m\hbar, (m=0, \pm 1, \pm 2, \dots)$

2. 归一化条件:

$\langle \psi_m | \psi_m \rangle = \int_0^{2\pi} \psi_m^* \psi_m d\varphi = 1$

$\therefore \int_0^{2\pi} |A|^2 d\varphi = 1 \Rightarrow A = \frac{1}{\sqrt{2\pi}}$

$\therefore \psi_m = \frac{1}{\sqrt{2\pi}} e^{im\varphi}$

本征方程: $\hat{L}_z \psi_m = m\hbar \psi_m$

(4) $\hat{L}^2$ 的本征值方程:

$$\begin{cases} \hat{L}^2 |Y\rangle = \lambda \hbar^2 |Y\rangle, \text{ 本征值: } \lambda \hbar^2. \\ \hat{L}^2 = -\hbar^2 \left[\frac{1}{\sin\theta} \frac{\partial}{\partial\theta} \left(\sin\theta \frac{\partial}{\partial\theta} \right) + \frac{1}{\sin^2\theta} \frac{\partial^2}{\partial\varphi^2} \right] \end{cases} \Rightarrow \text{球坐标系中: } \langle \hat{r} | \Psi \rangle = Y(\theta, \varphi)$$

联立得: $-\left\{ \frac{1}{\sin\theta} \frac{\partial}{\partial\theta} \left[\sin\theta \frac{\partial}{\partial\theta} Y(\theta, \varphi) \right] + \frac{1}{\sin^2\theta} \frac{\partial^2}{\partial\varphi^2} Y(\theta, \varphi) \right\} + \lambda Y(\theta, \varphi) = 0$

有限解:

$$\begin{cases} \lambda = \ell(\ell+1), \ell = 0, 1, 2, \dots \Rightarrow \text{决定 } L \text{ 的大小: 角量子数 } \ell. \\ m = 0, \pm 1, \pm 2, \dots, \pm\ell. \Rightarrow \text{决定 } L_z \text{ 的大小: 磁量子数 } m. \end{cases}$$

球函数: $Y_{\ell m}(\theta, \varphi) = (-1)^m N_{\ell m} P_{\ell}^m(\cos\theta) e^{im\varphi}$, $N_{\ell m} = \sqrt{\frac{2\ell+1}{4\pi} \frac{(\ell-m)!}{(\ell+m)!}}$ $\Rightarrow \hat{L}^2$ 与 $\hat{L}_z$ 的共同本征函数.

$= \Theta(\theta) \underline{\Phi(\varphi)}$ $\hookrightarrow \hat{L}_z$ 的本征函数.

$$\therefore \begin{cases} \hat{L}^2 |\ell m\rangle = \ell(\ell+1)\hbar^2 |\ell m\rangle \\ \hat{L}_z |\ell m\rangle = m\hbar |\ell m\rangle \end{cases}$$

二、力学量的共同本征态:

(一) 两个力学量的共同本征态: $|\varphi\rangle$

$$\hat{A}|\varphi\rangle = a|\varphi\rangle$$

$$\hat{B}|\varphi\rangle = b|\varphi\rangle$$

$$[\hat{A}, \hat{B}]|\varphi\rangle = (ab - ba)|\varphi\rangle = 0$$

$\Rightarrow$ 若 $\hat{A}, \hat{B}$ 对易, 则它们有共同本征态. (不可以不是共同本征态吗?)

Q. 不能推出 $[\hat{A}, \hat{B}] = 0$.

(二) 两个力学量的共同完备本征函数族.

定理: 两个力学量有共同完备的本征函数族的充要条件是这两个力学量对易. $\star$

自己证一遍?

三、不确定关系:

作业题里有用到过.

(一) 不确定度: 有一力学量 $\hat{A}$, 定义 $(\Delta A)^2 = \langle (\hat{A} - \bar{A})^2 \rangle = \bar{A^2} - \bar{A}^2$.

$$\therefore \hat{A} \text{ 的不确定度: } \Delta A = \sqrt{\bar{A^2} - \bar{A}^2}$$

(二) 不确定关系:

两条定理: ① 任意算符都可用厄米算符表示

② 任意算符 $\hat{F}$, 在任意态下 $\langle \psi | \hat{F}^\dagger \hat{F} | \psi \rangle \geq 0$ 即 $\langle \hat{F}^\dagger \hat{F} \rangle \geq 0$

要证:

(2). 不确定关系:

两个力学量 $\hat{A}, \hat{B}$. 定义: $\hat{F} = \hat{A} + i\lambda\hat{B}$. (λ 为实数). 则 $\hat{F}^\dagger = \hat{A} - i\lambda\hat{B}$.

$$\therefore \hat{F}^\dagger \hat{F} = \hat{A}^2 + i\lambda[\hat{A}, \hat{B}] + \lambda^2 \hat{B}^2$$

* 这里若 $\lambda=1$, 也可用均值不等式证出.

令 $[\hat{A}, \hat{B}] = i\hat{C}$. 其中 $\hat{C}$ 为厄米算符

$$\text{则: } \hat{F}^\dagger \hat{F} = \hat{A}^2 - \lambda^2 \hat{B}^2 + \lambda \hat{C}$$

又对任一力学量有: $\langle \hat{F}^\dagger \hat{F} \rangle \geq 0$

$$\therefore \langle \hat{A}^2 \rangle - \lambda^2 \langle \hat{B}^2 \rangle + \lambda \langle \hat{C} \rangle \geq 0$$

$$\Rightarrow \langle \hat{C} \rangle^2 - 4\lambda^2 \langle \hat{A}^2 \rangle \langle \hat{B}^2 \rangle \geq 0$$

$$\therefore \langle \hat{A}^2 \rangle \langle \hat{B}^2 \rangle \geq \frac{1}{4} \langle \hat{C} \rangle^2$$

而 $\hat{A} - \langle \hat{A} \rangle$ 和 $\hat{B} - \langle \hat{B} \rangle$ 也是厄米算符

$$\therefore \langle (\hat{A} - \langle \hat{A} \rangle)^2 \rangle \langle (\hat{B} - \langle \hat{B} \rangle)^2 \rangle \geq \frac{1}{4} \langle \hat{C} \rangle^2$$

$$(\Delta A)^2 (\Delta B)^2 \geq \frac{1}{4} \langle \hat{C} \rangle^2$$

$$\Rightarrow \Delta A \Delta B \geq \frac{1}{2} |\langle \hat{C} \rangle| \Rightarrow \text{若 } \hat{A}, \hat{B} \text{ 不对易, 则 } \langle \hat{C} \rangle \neq 0, A, B \text{ 不能同时取确定值}$$

例: 谐振子处于 ψ_0 态, 求 $\Delta x \Delta p = ?$

四. 力学量的平均值随时间变化:

(一). $\langle \hat{F} \rangle$ 随 t 变化:

$$\text{设 } \langle \hat{F} \rangle = \langle \psi | \hat{F} | \psi \rangle, \quad \frac{d}{dt} \langle \hat{F} \rangle = \frac{d}{dt} \langle \psi | \hat{F} | \psi \rangle = \langle \psi | \frac{d\hat{F}}{dt} | \psi \rangle + \langle \psi | \hat{F} \frac{d\psi}{dt} \rangle + \langle \frac{d\psi}{dt} | \hat{F} | \psi \rangle$$

$$\text{与薛方程联立: } \Rightarrow \text{Ehrenfest 定理: } \frac{d\langle \hat{F} \rangle}{dt} = \langle \frac{\partial \hat{F}}{\partial t} \rangle + \frac{1}{i\hbar} \langle [\hat{F}, \hat{H}] \rangle$$

? 举的两个例子没看懂

(二). 能量与 t 的不确定关系:

设任一力学量 $\hat{F}$.

$$[\hat{F}, \hat{H}] = i\hat{C}$$

$$\therefore \Delta F \Delta H \geq \frac{1}{2} |\langle \hat{C} \rangle|$$

$$\text{即 } \Delta F \Delta E \geq \frac{1}{2} |\langle \hat{C} \rangle|$$

$$\text{设 } \hat{F} \text{ 不显含时间, 由 Ehrenfest 定理: } \frac{d\langle \hat{F} \rangle}{dt} = \frac{1}{i\hbar} \langle [\hat{F}, \hat{H}] \rangle = \frac{1}{\hbar} \langle \hat{C} \rangle$$

$$\therefore \langle \hat{C} \rangle = \hbar \frac{d\langle \hat{F} \rangle}{dt}$$

$$\therefore \Delta F \Delta E \geq \frac{1}{2} \hbar \frac{d\langle \hat{F} \rangle}{dt}$$

$$\frac{\Delta F}{\frac{d\langle \hat{F} \rangle}{dt}} \Delta E \geq \frac{1}{2} \hbar \Rightarrow \Delta t \Delta E \geq \frac{1}{2} \hbar$$

(三) Hellmann - Feynman 定理. $\rightarrow$ 求均值好用: eg. $\bar{x}$, $\bar{p}$...

(11) 定理: 系统 $\hat{H}$ 不显含时间, ψ_n 为 $\hat{H}$ 的与 E_n 相对应的束缚本征态, $\hat{H}|\psi_n\rangle = E_n|\psi_n\rangle$, λ 为 $\hat{H}$ 的一个参数, 则: $\langle \psi_n | \frac{\partial \hat{H}}{\partial \lambda} | \psi_n \rangle = \frac{\partial E_n}{\partial \lambda}$ $\star$

回过头再证一下.

四. 对称性和守恒定律.

(12) 守恒量:

$$\star \begin{cases} \frac{d\langle \hat{F} \rangle}{dt} = 0 \\ [\hat{F}, \hat{H}] = 0 \end{cases} \Rightarrow \hat{F} \text{ 为守恒量} \Rightarrow \text{means: } \hat{F} \text{ 取不同本征值的概率不随时间变化.}$$

为什么守恒量 $\hat{F}$ 不随时间变化?

有一道例题:

(一). 对称性.

(1) 定义: 若一个变换能将系统从一个状态变换到另一个与之等价的状态, 则称这个变换为系统的对称变换, 或者说系统在变换下具有对称性. (变换下的不变性)

(2) 对称变换为么正变换: $\langle \psi | \psi \rangle = 1$

设 $\hat{U}$ 表示变换: $\hat{U}|\psi\rangle = |\psi'\rangle$

$$\therefore \langle \psi | \hat{U}^\dagger = \langle \psi' |$$

$$\text{又 } \langle \psi' | \psi' \rangle = 1$$

$$\therefore \langle \psi | \hat{U}^\dagger \hat{U} | \psi \rangle = 1 = \langle \psi | \psi \rangle$$

$$\therefore \hat{U}^\dagger \hat{U} = 1$$

$$\therefore \hat{U}^\dagger = \hat{U}^{-1}$$

(3) 力学量的变换: $\hat{F}' = \hat{F}$

$$\therefore \bar{F} = \langle \psi | \hat{F} | \psi \rangle = \langle \psi' | \hat{F}' | \psi' \rangle$$

$$= \langle \psi | \hat{U}^\dagger \hat{F}' \hat{U} | \psi \rangle$$

$$\therefore \hat{F} = \hat{U}^\dagger \hat{F}' \hat{U} = \hat{U}^\dagger \hat{F}' \hat{U} \text{ (相似变换)}$$

对哈密顿量: $\hat{H}' = \hat{U}^\dagger \hat{H} \hat{U}$

又变换下能量不变: $\therefore \bar{H} = \bar{H}'$

$$\therefore \bar{H} = \langle \psi | \hat{H} | \psi \rangle \Rightarrow \langle \psi | \hat{H} - \hat{H}' | \psi \rangle = \langle \psi | \hat{U}^\dagger \hat{H} \hat{U} - \hat{H} | \psi \rangle = 0$$

$\downarrow$
意味着变换 $\hat{U}$ 不改变系统的能量.

$$\Rightarrow \begin{cases} \hat{F}' = \hat{U} \hat{F} \hat{U}^\dagger \\ [\hat{U}, \hat{H}] = 0 \\ \hat{U}^\dagger \hat{U} = 1 \end{cases} \Rightarrow \hat{U} \text{ 为对称变换.} \star$$

(三). 无穷小对称变换:

$$\hat{U} = I + \hat{T} = I + i\varepsilon \hat{F} \quad \varepsilon \text{ 为无穷小}$$

$$= e^{i\varepsilon \hat{F}}$$

$$\text{则: } \hat{U}^\dagger = I - i\varepsilon \hat{F}^\dagger = e^{-i\varepsilon \hat{F}^\dagger}$$

若 $\hat{U}$ 为对称变换算符, 则有: $\hat{U}^\dagger \hat{U} = I$

$$\therefore (I - i\varepsilon \hat{F}^\dagger)(I + i\varepsilon \hat{F}) = I^2 + i\varepsilon(\hat{F} - \hat{F}^\dagger) + \varepsilon^2 \hat{F}^\dagger \hat{F} \approx I + i\varepsilon(\hat{F} - \hat{F}^\dagger) = e^{i\varepsilon(\hat{F} - \hat{F}^\dagger)} = I \Rightarrow \hat{F} = \hat{F}^\dagger$$

$\hat{F}$ 为 $\hat{U}$ 的生成元, 即无穷小变换可由力学量表示: $[\hat{U}, \hat{H}] = 0 \Rightarrow [\hat{F}, \hat{H}] = 0$

诺特定理: 系统在连续变换下的不变性, 对应着必有一个力学量守恒.

(四). 空间平移不变性.

设 $\hat{U}_0$ 表示平移变换: $\therefore |4\rangle \rightarrow |4'\rangle = \hat{U}_0 |4\rangle$

平移: $\vec{r} \rightarrow \vec{r}' = \vec{r} + \delta \vec{r}$

平移不变即: $\psi(\vec{r}') = \psi(\vec{r})$

$$\therefore \psi(\vec{r} + \delta \vec{r}) = \psi(\vec{r})$$

$$\therefore \psi(\vec{r}) = \psi(\vec{r} - \delta \vec{r})$$

↓

- 维: $\psi(x) = \psi(x - \delta x)$

$$= \sum_{n=0}^{\infty} \frac{\partial^n}{\partial x^n} \psi(x) \frac{(-\delta x)^n}{n!}$$

$$= \sum_{n=0}^{\infty} \frac{(-\delta x)^n}{n!} \frac{\partial^n}{\partial x^n} \psi(x)$$

$$= \boxed{e^{-\delta x \frac{\partial}{\partial x}}} \psi(x)$$

$$\text{而: } |4'\rangle = \hat{U}_0 |4\rangle \quad \therefore \hat{U}_0 = e^{-\delta x \frac{\partial}{\partial x}} = e^{-\frac{i}{\hbar} \delta x \hat{p}_x}$$

$$\Rightarrow \text{三维: } \hat{U}_0 = e^{-\frac{i}{\hbar} \delta \vec{r} \cdot \vec{p}}$$

$$\therefore \text{平移不变: } [\hat{U}_0, \hat{H}] = 0 \Rightarrow [\vec{p}, \hat{H}] = 0$$

↓
means: 角动量守恒.

(五). 时间平移不变性.

设 $\hat{U}_T$ 表示时间平移变换:

$$\text{同理可得: } \hat{U}_T = e^{-i\hat{H}\delta t} = \text{该算符与 } \hat{H}$$

$$\therefore \text{时间平移不变: } [\hat{U}_T, \hat{H}] = 0$$

$$\Rightarrow \frac{\partial \hat{H}}{\partial t} = 0 \quad \therefore \text{能量守恒.}$$

五. 空间转动不变性.

对无穷小转动: $\vec{r} \rightarrow \vec{r}' = \hat{T} \vec{r}$

$$\begin{cases} x' = x + \epsilon_{11}x + \epsilon_{12}y + \epsilon_{13}z \\ y' = y + \epsilon_{21}x + \epsilon_{22}y + \epsilon_{23}z \\ z' = z + \epsilon_{31}x + \epsilon_{32}y + \epsilon_{33}z \end{cases}$$

$$\therefore \hat{T} = I + \hat{\epsilon}$$

$$\therefore \hat{T}^\dagger = I + \hat{\epsilon}^\dagger$$

$$\therefore \hat{T}^\dagger \hat{T} = I \Rightarrow I^2 + \hat{\epsilon} + \hat{\epsilon}^\dagger + \hat{\epsilon} \hat{\epsilon}^\dagger \approx 2I^2 + \hat{\epsilon} + \hat{\epsilon}^\dagger = I$$

$$\Rightarrow \hat{\epsilon}^\dagger = -\hat{\epsilon}$$

$$\therefore \epsilon_{ij} = -\epsilon_{ji}$$

$$\text{令 } \hat{\epsilon} = \begin{pmatrix} 0 & -\delta\varphi_z & \delta\varphi_y \\ \delta\varphi_z & 0 & -\delta\varphi_x \\ -\delta\varphi_y & \delta\varphi_x & 0 \end{pmatrix} \Rightarrow \begin{cases} x' = x - \delta\varphi_z y + \delta\varphi_y z \\ y' = y + \delta\varphi_z x - \delta\varphi_x z \\ z' = z + \delta\varphi_y x + \delta\varphi_x y \end{cases} \Rightarrow \vec{r}' = \vec{r} + \delta\vec{\varphi} \times \vec{r}$$

$$\text{定义: } \delta\vec{\varphi} = \delta\varphi_x \vec{e}_x + \delta\varphi_y \vec{e}_y + \delta\varphi_z \vec{e}_z \\ = \delta\varphi \vec{n}$$

$$\psi(\vec{r}') = \psi(\vec{r})$$

$$\therefore \psi(\vec{r} + \delta\vec{\varphi} \times \vec{r}) = \psi(\vec{r})$$

$$\therefore \psi(\vec{r}) = \psi(\vec{r} - \delta\vec{\varphi} \times \vec{r})$$

$$= \psi(\vec{r}) - (\delta\vec{\varphi} \times \vec{r}) \cdot \vec{\nabla} \psi(\vec{r}) + \dots$$

$$\approx \psi(\vec{r}) - \delta\vec{\varphi} \cdot \left(\frac{\vec{r} \times \vec{\nabla}}{\hbar} \right) \psi(\vec{r})$$

$$= \psi(\vec{r}) - \frac{i}{\hbar} \delta\vec{\varphi} \cdot (\vec{r} \times \hat{p}) \psi(\vec{r})$$

$$= \psi(\vec{r}) - \frac{i}{\hbar} \delta\vec{\varphi} \cdot \hat{L} \psi(\vec{r})$$

$$= \psi(\vec{r}) - \frac{i}{\hbar} \delta\varphi \hat{L}_n \psi(\vec{r})$$

$$= e^{-\frac{i}{\hbar} \delta\varphi \hat{L}_n} \psi(\vec{r})$$

$$\text{转动算符: } \hat{U}_n = e^{-\frac{i}{\hbar} \delta\varphi \hat{L}_n}$$

$$\therefore \text{转动对称性: } [\hat{U}_n, \hat{H}] = 0 \Rightarrow [\hat{L}_n, \hat{H}] = 0$$

若 $\vec{n}$ 是任意的, 则角动量 $\hat{L}$ 守恒.

六、空间反演

$$\vec{r} \rightarrow \vec{r}' = -\vec{r}, \quad \vec{p} \rightarrow \vec{p}' = -\vec{p}, \quad \therefore \vec{L}' = \vec{r}' \times \vec{p}' = \vec{L}$$

$$\text{镜象变换} \begin{cases} \text{矢量} \begin{cases} \text{平行镜面的分量方向不变} \\ \text{垂直变号} \end{cases} \\ \text{赝矢量} \begin{cases} \text{平行变号} \\ \text{垂直不变} \end{cases} \end{cases}$$

$\Rightarrow$ 可用于记 $\vec{E}, \vec{B}$ 的边界条件.

设 $\hat{\pi}$ 为空间反演算符:

$$\text{有: } \psi(\vec{r}') = \hat{\pi} \psi(\vec{r})$$

$$\psi(-\vec{r}) = \hat{\pi} \psi(\vec{r})$$

$$\therefore \psi'(\vec{r}) = \hat{\pi} \psi(-\vec{r}) = \pi \psi(\vec{r})$$

$$\therefore \hat{\pi} \psi(\vec{r}) = \psi(-\vec{r})$$

$$\hat{\pi}^2 \psi(\vec{r}) = \psi(\vec{r})$$

$$\therefore \hat{\pi}^2 = 1, \text{ 即 } \hat{\pi} \hat{\pi} = 1$$

又 $\hat{\pi}^+ \hat{\pi} = 1$ 即 $\hat{\pi} = \hat{\pi}^+$ 为厄米算符. $\Rightarrow \hat{\pi}$ 为一力学量, 定义为: 宇称.

描述物理规律在镜像对称下是否守恒.

$[\hat{\pi}, \hat{H}] = 0$. 即空间反演不变说明宇称守恒.

若宇称变化: $\hat{\pi} u(\vec{r}) = \lambda u(\vec{r})$

$$\therefore \hat{\pi}^2 u(\vec{r}) = \lambda^2 u(\vec{r}) \Rightarrow \lambda^2 = 1 \quad \left\{ \begin{array}{l} \lambda = 1: \text{偶宇称态} \quad \psi(-\vec{r}) = \psi(\vec{r}) \\ \lambda = -1: \text{奇宇称态} \quad \psi(-\vec{r}) = -\psi(\vec{r}) \end{array} \right.$$

(原子, 核, 电子)

(光子, 介子)

什么叫宇称变化? $\hat{\pi} u(\vec{r}) = \lambda u(\vec{r})$ 是为什么?

第七章. 中心力场的本征值问题.

一. 中心力场的一般描述: $V = V(r)$

(一). 定态薛定谔方程:

$$\hat{H}\psi(\vec{r}) = E\psi(\vec{r}) \quad \dots \textcircled{1} \quad (\text{坐标表象下})$$

$$\hat{H} = -\frac{\hbar^2}{2\mu}\nabla^2 + V(r) \quad \dots \textcircled{2}$$

$$\text{球坐标系中: } \nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2\frac{\partial}{\partial r}) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}(\sin\theta\frac{\partial}{\partial\theta}) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial\varphi^2}$$

$$= \frac{1}{r^2}\frac{\partial}{\partial r}(r^2\frac{\partial}{\partial r}) - \frac{1}{r^2}\hat{L}^2 \quad \dots \textcircled{3}$$

$$\therefore \hat{H} = -\frac{\hbar^2}{2\mu r^2}\frac{\partial}{\partial r}(r^2\frac{\partial}{\partial r}) + \frac{1}{2\mu r^2}\hat{L}^2 + V(r) \quad \dots \textcircled{4}$$

将④代入①:

$$[-\frac{\hbar^2}{2\mu r^2}\frac{\partial}{\partial r}(r^2\frac{\partial}{\partial r}) + \frac{1}{2\mu r^2}\hat{L}^2 + V(r)]\psi(\vec{r}) = E\psi(\vec{r})$$

$$[-\frac{\hbar^2}{2\mu r^2}\frac{\partial}{\partial r}(r^2\frac{\partial}{\partial r}) + (V(r) - E)]\psi(\vec{r}) = -\frac{1}{2\mu r^2}\hat{L}^2\psi(\vec{r})$$

$$\therefore [\frac{\partial}{\partial r}(r^2\frac{\partial}{\partial r}) + \frac{2\mu r^2}{\hbar^2}(E - V(r))]\psi(\vec{r}) = \frac{1}{\hbar^2}\hat{L}^2\psi(\vec{r}) \\ = -[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}(\sin\theta\frac{\partial}{\partial\theta}) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\varphi^2}]\psi(\vec{r}) \quad \dots \textcircled{5}$$

分离变量, $\psi(\vec{r}) = R(r)Y(\theta, \varphi)$. 代入⑤式:

$$\frac{\partial}{\partial r}(r^2\frac{\partial}{\partial r}R(r))Y(\theta, \varphi) + \frac{2\mu r^2}{\hbar^2}(E - V(r))R(r)Y(\theta, \varphi) = -[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}(\sin\theta\frac{\partial}{\partial\theta}Y(\theta, \varphi)) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\varphi^2}Y(\theta, \varphi)]R(r)$$

两边同除 $R(r)Y(\theta, \varphi)$:

$$\frac{1}{R(r)}\frac{\partial}{\partial r}(r^2\frac{\partial}{\partial r}R(r)) + \frac{2\mu r^2}{\hbar^2}(E - V(r)) = -\frac{1}{Y}[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}(\sin\theta\frac{\partial}{\partial\theta}Y) + \frac{1}{\sin^2\theta}\frac{\partial^2 Y}{\partial\varphi^2}] = \lambda \quad \rightarrow \text{与方程中变量无关.}$$

$$\Rightarrow \begin{cases} -[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}(\sin\theta\frac{\partial}{\partial\theta}Y) + \frac{1}{\sin^2\theta}\frac{\partial^2 Y}{\partial\varphi^2}] = \lambda Y & \Rightarrow \text{第1章已解过: } \hat{L}^2 Y = \lambda \hbar^2 Y \Rightarrow Y = (-1)^m N_{lm} P_l^m(\cos\theta) e^{im\varphi} \\ \frac{1}{r^2}\frac{\partial}{\partial r}(r^2\frac{\partial}{\partial r}R) + \frac{2\mu}{\hbar^2}(E - V(r))R = \frac{\lambda}{r^2}R & \quad (\lambda = l(l+1)\hbar^2) \quad \leftarrow \text{与 } \hbar^2 \text{ 约去} \end{cases}$$

$l=0,1,2, \dots$
 $m=0, \pm 1, \pm 2, \dots$
 $\lambda = l(l+1)$

(二). 径向薛定谔方程. (核心求解方程)

$\lambda = l(l+1)$. ⑥式可改写为:

$$-\frac{\hbar^2}{2\mu} \frac{1}{r} \frac{d}{dr}(r^2 \frac{dR}{dr}) + (V + \frac{l(l+1)\hbar^2}{2\mu r^2})R = ER \quad \dots \textcircled{7}$$

中心力场中的经典哈密顿量:

$$\hat{H} = \frac{\vec{p}^2}{2\mu} + V(r)$$

$$= \frac{p_r^2}{2\mu} + \frac{\vec{p}_\perp^2}{2\mu} + V(r) \Rightarrow \text{径向动量: } p_r = \frac{\vec{r}}{r} \cdot \vec{p}, \text{ 横向动量: } \vec{p}_\perp = \frac{\vec{r}}{r} \times \vec{p} = \frac{\vec{L}}{r}$$

$$= \frac{p_r^2}{2\mu} + \left[\frac{\vec{L}^2}{2\mu r^2} + V(r) \right] \Rightarrow U(r) = V(r) + \frac{\vec{L}^2}{2\mu r^2} \text{ 有效势能.}$$

②式可改写为

而在量子力学中, $\hat{H}$ 表示为:
$$\hat{H} = -\frac{\hbar^2}{2\mu r^2} \frac{d}{dr} \left(r^2 \frac{d}{dr} \right) + \frac{\hat{L}^2}{2\mu r^2} + V(r)$$
$$= \frac{p_r^2}{2\mu} + \frac{\hat{L}^2}{2\mu r^2} + V(r).$$

对比知: $p_r^2 = -\frac{\hbar^2}{r^2} \frac{d}{dr} \left(r^2 \frac{d}{dr} \right)$ ($\hat{p}_r$ 不是厄米算符)

利用玻姆规则可构造出: $\hat{p}_r = \frac{1}{2} \left(\frac{\vec{r} \cdot \vec{p}}{r} + \frac{\vec{p} \cdot \vec{r}}{r} \right) = -i\hbar \left(\frac{\partial}{\partial r} + \frac{1}{r} \right)$ $\rightarrow$ 易证: $p_r^2 = -\frac{\hbar^2}{r^2} \frac{d}{dr} \left(r^2 \frac{d}{dr} \right)$

$[r, \hat{p}_r] = [r, -i\hbar \left(\frac{\partial}{\partial r} + \frac{1}{r} \right)] = i\hbar$

②式可改写为: $\frac{\hbar^2}{2\mu} \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + (E - V - \frac{\mu(u+1)\hbar^2}{2\mu r^2}) R = 0$; 也即: $\frac{\hat{p}_r^2}{2\mu} R + \left(\frac{\hat{L}^2}{2\mu r^2} + V \right) R = ER$ 好记!!!

令 $U(r) = r R(r)$

☆ $\frac{\hbar^2}{2\mu} \frac{d^2}{dr^2} U(r) + [E - (V(r) + \frac{\mu(u+1)\hbar^2}{2\mu r^2})] U(r) = 0 \Rightarrow$ 径向薛定谔方程.

即: $-\frac{\hbar^2}{2\mu} \frac{d^2}{dr^2} U(r) + U(r) U(r) = E U(r)$, $U(r) = V(r) + \frac{\mu(u+1)\hbar^2}{2\mu r^2} \Rightarrow$ 与一维定态薛方程形式相同

与 m 无关, 显现出球对称势场的特点. 能级与 m 无关, 对 m 简并, 简并度为 $2l+1$ (不考虑自旋)

下面对是否有解, 是否有稳定的束缚态解进行讨论:

设 $V = -\frac{A}{r^s}$, $r \rightarrow \infty, V \rightarrow 0, s > 0$ $\left\{ \begin{array}{l} A > 0, \text{ 吸引势} \\ A < 0, \text{ 排斥势} \end{array} \right.$

(1). $E > 0, r \rightarrow \infty, U \rightarrow 0$ (非束缚态, 电离)

$\therefore$ 径向薛: $\frac{d^2}{dr^2} U(r) + \frac{2\mu E}{\hbar^2} = 0 \Rightarrow U(r)$ 为振荡解, 即平面波. $\therefore R(r) = \frac{U(r)}{r}$ 为球面波.

(2). $E < 0, r \rightarrow \infty, U \rightarrow 0$

$\therefore$ 径向薛: $\frac{d^2}{dr^2} U(r) + \frac{2\mu E}{\hbar^2} = 0$

若要存在 $E < 0$ 的解, 则 $A > 0$ 为吸引势

$\therefore k^2 = -\frac{2\mu E}{\hbar^2}$ $U(r) = ae^{kr} + be^{-kr} = be^{-kr}$ (束缚态, 能级分立).

(3). $U(r) = -\frac{A}{r^s} + \frac{\mu(u+1)\hbar^2}{2\mu r^2}$, $s > 0, A > 0$

①. $s > 2$ 时: $r \rightarrow 0, U \sim -\frac{A}{r^s}$ 吸引势 $\left. \begin{array}{l} r \rightarrow \infty, U \sim \frac{B}{r^2} \text{ 排斥势} \end{array} \right\}$ 不存在稳定的束缚态解.

②. $0 < s < 2$ 时: $r \rightarrow 0, U \sim \frac{B}{r^2}$ 排斥势 $\left. \begin{array}{l} r \rightarrow \infty, U \sim -\frac{A}{r^s} \text{ 吸引势} \end{array} \right\}$ 稳定的束缚态解可以存在. ✓

③. $s = 2$ 时: $-\frac{\hbar^2}{2\mu} U''(r) + \frac{B-A}{r^2} U(r) = EU(r)$

$\left\{ \begin{array}{l} A > \frac{\hbar^2}{2\mu} \text{ 坍塌} \\ A < \frac{\hbar^2}{2\mu} \text{ 不坍塌} \end{array} \right.$

(三). 原子(电子)的磁矩(轨道磁矩)

1. 经典理论



$$d\vec{A} = \frac{1}{2} \vec{r} \times d\vec{r}$$

$$\therefore \frac{d\vec{A}}{dt} = \frac{1}{2} \vec{r} \times \vec{v} = \frac{1}{2\mu} \vec{r} \times \vec{p} = \frac{\vec{L}}{2\mu} \quad (\text{掠面速度})$$

$$\vec{A} = \oint d\vec{A} = \oint \frac{\vec{L}}{2\mu} dt = \frac{\vec{L}}{2\mu}$$

$$\text{轨道磁矩: } \vec{\mu}_L = I\vec{A}, \quad I = -\frac{e}{T}$$

$$\therefore \vec{\mu}_L = -\frac{e}{2\mu} \vec{L} = \gamma \vec{L} \quad \begin{cases} L = \sqrt{l(l+1)} \hbar \\ \gamma = \frac{e}{2\mu} \text{ 旋磁比} \end{cases}$$

$$\therefore \mu_{Lz} = -\frac{e}{2\mu} L_z = -M \frac{e\hbar}{2\mu} = -m\mu_B, \quad \mu_B = \frac{e\hbar}{2\mu} \text{ 玻尔磁子.}$$

2. 量子理论:

$$\text{概率流密度: } \vec{j}(\vec{r}, t) = \frac{i\hbar}{2\mu} (\psi \vec{\nabla} \psi^* - \psi^* \vec{\nabla} \psi), \quad \psi(\vec{r}) = R(r) Y(\theta, \varphi).$$

$$\vec{\nabla} = \frac{\partial}{\partial r} \vec{e}_r + \frac{1}{r} \frac{\partial}{\partial \theta} \vec{e}_\theta + \frac{1}{r \sin \theta} \frac{\partial}{\partial \varphi} \vec{e}_\varphi$$

$\therefore$ 径向波函数, $P_n^m(\cos \theta)$ 均为实数.

$$\therefore j_r = \frac{i\hbar}{2\mu} (\psi \frac{\partial}{\partial r} \psi^* - \psi^* \frac{\partial}{\partial r} \psi) = 0$$

$$j_\theta = \frac{i\hbar}{2\mu} (\psi \frac{1}{r} \frac{\partial}{\partial \theta} \psi^* - \psi^* \frac{1}{r} \frac{\partial}{\partial \theta} \psi) = 0$$

$$j_\varphi = \frac{i\hbar}{2\mu} \frac{1}{r \sin \theta} (\psi \frac{\partial}{\partial \varphi} \psi^* - \psi^* \frac{\partial}{\partial \varphi} \psi) = \frac{m\hbar}{\mu r \sin \theta} \psi^* \psi$$

$$\therefore \vec{j}_e = -e \vec{j}(\vec{r}, t).$$

$$j_{e\varphi} = -e j_\varphi = -\frac{em\hbar}{\mu r \sin \theta} \psi^* \psi. \quad (\text{安培分子环流的理论依据})$$

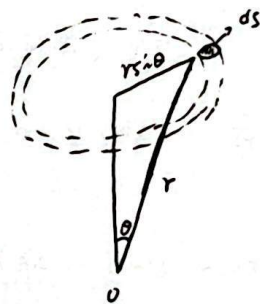
$$ds = r dr d\theta$$

$$dI = j_{e\varphi} ds = j_{e\varphi} r dr d\theta$$

$$\therefore d\mu_z = A dI = \lambda (r \sin \theta)^2 dI.$$

$$\therefore \mu_z = \int d\mu_z = \int -\lambda (r \sin \theta)^2 \frac{em\hbar}{\mu r \sin \theta} \psi^* \psi dr d\theta = -m \frac{e\hbar}{2\mu} \int \psi^* \psi d\tau.$$

$$\Rightarrow \mu_z = -m\mu_B = -m \frac{e\hbar}{2\mu}$$



二. 氢原子与类氢原子.

"两体问题" $\xrightarrow{\text{质心}}$ "单体问题"

相对运动有: $\mu = \frac{Mm}{M+m}$, $\vec{r} = \vec{r}_2 - \vec{r}_1$.

径向薛定谔方程有: $\frac{d^2}{dr^2} u(r) + \frac{2\mu}{\hbar^2} [E - (V(r) + \frac{\lambda(\lambda+1)\hbar^2}{2\mu r^2})] u(r) = 0$

库仑势: $V(r) = -\frac{Ze^2}{4\pi\epsilon_0 r}$ 氢原子 $Z=1$, 类氢原子 $Z>1$.

$$\therefore \frac{d^2}{dr^2} u(r) + \frac{2\mu}{\hbar^2} [E + (\frac{Ze^2}{4\pi\epsilon_0 r} - \frac{\lambda(\lambda+1)\hbar^2}{2\mu r^2})] u(r) = 0$$

束缚解条件: $u(r) \xrightarrow{r \rightarrow 0} \text{有限}$, $u(r) \xrightarrow{r \rightarrow \infty} 0$.

①. $E > 0$. 电离态, 能量取连续值.

$$\text{②. } E < 0. \quad 1^\circ. r \rightarrow 0, \quad \frac{d^2}{dr^2} u(r) - \frac{\lambda(\lambda+1)}{r^2} u(r) = 0$$

$$\Rightarrow u(r) = A r^{\lambda+1} + B r^{-\lambda}. \quad \because r \rightarrow 0, u(r) \text{ 有限.}$$

$$\therefore u(r) = C r^{\lambda+1}$$

$$2^\circ. r \rightarrow \infty, \quad \frac{d^2}{dr^2} u(r) + \frac{2\mu E}{\hbar^2} u(r) = 0.$$

$$\text{令 } k^2 = \frac{-2\mu E}{\hbar^2} \quad \therefore u(r) = A e^{kr} + B e^{-kr}$$

$$\because r \rightarrow \infty, u(r) \rightarrow 0.$$

$$\therefore u(r) = C e^{-kr}.$$

故设 $u(r) = C r^{\lambda+1} F(r) e^{-kr}$, 令 $\xi = 2kr$.

$$\text{方程改写为: } \xi \frac{d^2 F}{d\xi^2} + (2\lambda + 2 - \xi) \frac{dF}{d\xi} - (\lambda + 1 - \frac{Ze^2}{4\pi\epsilon_0 \hbar^2 \sqrt{-2\mu E}}) F = 0$$

合流超几何方程: $\xi f''(x) + (b-x)f'(x) - af = 0$

$$\text{解得: } \star \quad \underline{E_n = -\frac{1}{2} \frac{\mu Z^2 e^4}{(4\pi\epsilon_0 \hbar)^2}} \quad (n=1, 2, 3, \dots)$$

$$\underline{R_{nl}(r) = \frac{u_{nl}(r)}{r} = N_{nl} \rho^\lambda e^{-\frac{\rho}{2}} L_{n-\lambda-1}^{2\lambda+1}(\rho)}. \quad \lambda = 0, 1, 2, \dots, n-1$$

其中: $\rho = \frac{2Zr}{na_0} = \alpha r$, $a_0 = \frac{4\pi\epsilon_0 \hbar^2}{\mu e^2} = 0.0529 \text{ nm}$. 基态时玻尔轨道半径.

$$L_n^k(x) = \frac{x^{-k} e^x}{n!} \frac{d^n}{dx^n} (x^{n+k} e^{-x}) = \sum_{m=0}^n (-1)^m C_{n+k}^{n-m} \frac{x^m}{m!} \quad \text{连带拉盖尔多项式}$$

$$N_{nl} = \left(\frac{2Z}{na_0} \right)^3 \frac{(n-\lambda-1)!}{2^n [(n+1)!]^{1/2}}$$

讨论 ①. 能量不能连续取值 $\rightarrow$ 自然给出了玻尔的氢原子能级公式, 无需其它假定.

②. 不能级不依赖 m 也不依赖 λ . 库仑场的特征.

库仑场 (~ $1/r$) 较一般中心力场有更高的对称性 $\rightarrow$ 对称性 / 简并度

$$\text{③. } \psi_{nlm}(-\vec{r}) = (-1)^l \psi_{nlm}(\vec{r}) : \quad \lambda \text{ 宇称.}$$

由角动量守恒: $[\hat{L}_z, \hat{H}] = 0 \Rightarrow [\hat{L}^2, \hat{H}] = 0$
且 $[\hat{L}_z, \hat{L}^2] = 0$

$$\begin{cases} \hat{H} \psi_{nlm}(\vec{r}) = E_n \psi_{nlm}(\vec{r}) & E_n = -\frac{1}{2} \frac{\mu e^4}{(4\pi\epsilon_0 \hbar)^2} \frac{1}{n^2}, n=0, 1, 2, 3 \dots \text{主量子数} \rightarrow \text{决定能级大小} \\ \hat{L}^2 \psi_{nlm}(\vec{r}) = \hbar^2 l(l+1) \psi_{nlm}(\vec{r}) & \hat{L}^2 = \hbar^2 l(l+1), l=0, 1, 2 \dots n-1 \text{角量子数} \rightarrow \text{决定角动量的大小} \\ \hat{L}_z \psi_{nlm}(\vec{r}) = m \hbar \psi_{nlm}(\vec{r}) & L_z = m \hbar, m=0, \pm 1, \pm 2 \dots \pm l \text{磁量子数} \rightarrow \text{决定角动量投影分量的大小} \end{cases}$$

能级简并度: $f_n = \sum_{l=0}^{n-1} (2l+1) = n^2$ 考虑自旋 $f_n = 2n^2$

讨论: 在 $\psi_{nlm}(\vec{r}) = R_{nl}(r) Y_{lm}(\theta, \varphi)$ 态下电子的概率分布

$$W_{nlm}(r, \theta, \varphi) d\tau = |\psi_{nlm}(\vec{r})|^2 r^2 \sin\theta dr d\theta d\varphi$$

在 $r \sim r+dr$ 内找到粒子的概率: $W(r) dr = |R_{nl}(r)|^2 r^2 dr \int |Y_{lm}(\theta, \varphi)|^2 d\Omega$

$$Q = R_{nl}^2(r) r^2 dr$$

∴ 概率密度: $w_{nl}(r) = r^2 R_{nl}^2(r)$

eg. $w_{10}(r) = r^2 R_{10}^2(r) = \frac{4r^2}{a_0^3} e^{-\frac{2r}{a_0}}, R_{10}(r) = \left(\frac{2}{a_0}\right)^{\frac{1}{2}} e^{-\frac{r}{a_0}}$ 氢原子: $z=1$

① 最概然半径: $\frac{d}{dr} w_{10}(r) = 0 \Rightarrow r = a_0$

当 $l = n-1$ 时: $\Rightarrow r = n^2 a_0$

② 平均半径: $\langle r \rangle = \int \psi_{nlm}^* r \psi_{nlm} d\tau = \int_0^\infty r^3 R_{nl}^2(r) dr$
 $= \frac{1}{2} [3n^2 - l(l+1)] a_0$

当 $l = n-1$ 时, $\langle r \rangle = \left(\frac{n}{2} + n^2\right) a_0$

③ 径向零点的零点: 不考虑 $r \rightarrow 0$ 和 $r \rightarrow \infty$ 两处

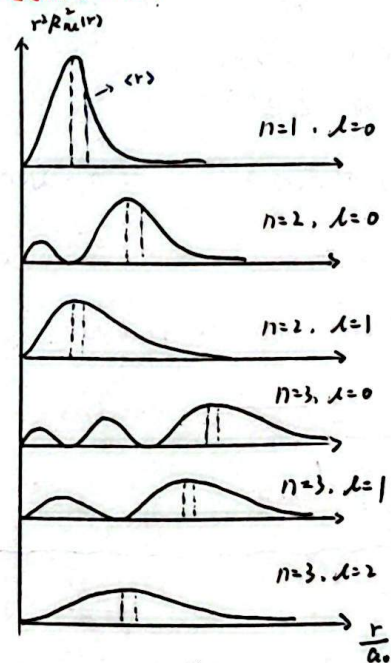
$R_{nl}(r)$ 共有 $n-l-1$ 个零点

$n-l$ 个极值点

④ 碱金属原子的价电子: $E_n = -\frac{1}{2} \frac{Z^* \mu e^4}{(4\pi\epsilon_0 \hbar)^2} \frac{1}{n^2} \Rightarrow E_n$

能级分裂成 n 条 (Z^* 与 l 有关, $l=0, 1, 2 \dots n-1$) $\rightarrow 3s, 3p, 3d \dots$ (但氢原子没有)

⑤ 概率的角分布: 对 r 积分得: $w_{lm}(\theta, \varphi) = |Y_{lm}(\theta, \varphi)|^2 d\Omega \Rightarrow$ 与 φ, n 无关, 轴对称



PPT上例题: 求 $\langle \frac{1}{r} \rangle$? $n = k+l+1$ 是为什么?

第八章. 表象理论

一. 状态的表象

(1). 力学量: $\hat{F}^\dagger = \hat{F}$

$\hat{F}|u_n\rangle = f_n|u_n\rangle$: $\hat{F}$ 的本征矢量 $\{|u_n\rangle\}$ 构成完备的本征函数族. $\rightarrow$ 张成的希尔伯特空间称为 " $\hat{F}$ 表象"

完备性的数学表述: $\sum_n |u_n\rangle \langle u_n| = 1$.

$\therefore \psi$ 有: $|\psi\rangle = \sum_n C_n |u_n\rangle$, $C_n = \langle u_n | \psi \rangle$, 也即 取某态对应的本征值的概率

则在 F 表象下, ψ 可写为 - 列矩阵: $\psi = \begin{pmatrix} C_1 \\ C_2 \\ \vdots \\ C_n \end{pmatrix} = \begin{pmatrix} \langle u_1 | \psi \rangle \\ \langle u_2 | \psi \rangle \\ \vdots \\ \langle u_n | \psi \rangle \end{pmatrix} \Rightarrow$ 前提是表象确定.

e.g. 坐标表象下: $\psi = \begin{pmatrix} \psi(x) \end{pmatrix} \rightarrow \psi(x)$ (仅为 - 矩阵元).

(2). 对连续谱:

$\hat{F}|u_f\rangle = f|u_f\rangle$, f 连续取值.

$$\begin{aligned} \therefore |\psi\rangle &= \int C(f) |u_f\rangle df \\ &= \int C(f) |u_f\rangle df \\ &= \int \langle u_f | \psi \rangle \langle u_f \rangle df \end{aligned}$$

在坐标表象下: $\langle r | \psi \rangle = \langle r | \cdot \int C(f) |u_f\rangle df$

$$\begin{aligned} \psi(r, t) &= \int C(f, t) \langle r | u_f \rangle df \\ &= \int C(f, t) \langle r | u_f \rangle df \end{aligned}$$

$$\begin{aligned} \therefore C(f, t) &= \langle u_f | \psi \rangle \\ &= \int \langle u_f | r \rangle \langle r | \psi \rangle dr \\ &= \int u_f^*(r) \psi(r, t) dr \end{aligned}$$

(3). $\hat{H}$ 表象. ($\hat{H}$ 不含 t)

讨论 $|\psi\rangle = \hat{H}|\psi\rangle$

$$\begin{aligned} \therefore \ln |\psi\rangle \Big|_0^t &= -\frac{i}{\hbar} \hat{H} t \Big|_0^t \\ \Rightarrow |\psi(t)\rangle &= e^{-\frac{i}{\hbar} \hat{H} t} |\psi(0)\rangle \end{aligned}$$

又: $\hat{H}|\varphi_n\rangle = E_n |\varphi_n\rangle$

$$\therefore |\psi(0)\rangle = \sum_n C_n |\varphi_n\rangle \quad \star$$

$$\begin{aligned} \therefore \underline{|\psi(t)\rangle} &= e^{-\frac{i}{\hbar} \hat{H} t} \sum_n C_n |\varphi_n\rangle \\ &= \sum_n C_n e^{-\frac{i}{\hbar} E_n t} |\varphi_n\rangle \\ &= \underline{\sum_n C_n e^{-\frac{i}{\hbar} E_n t} |\varphi_n\rangle} \quad \star \end{aligned}$$

自身

只有态矢量用 $\hat{H}$ 表象下展开才能乘时间因子 $e^{-\frac{i}{\hbar} \hat{H} t}$ $\star$

或用自己的本征矢展开

二. 力学量的算符表示 (矩阵表示)

(一). 力学量的表示:

$\hat{F}|u_n\rangle = f_n|u_n\rangle \quad \{|u_n\rangle\}$. 力学量 $\hat{Q}$ 在 $\hat{F}$ 表象下如何表示?

$$\psi|4\rangle: \hat{Q}|4\rangle = |\varphi\rangle.$$

$$\langle u_m|\hat{Q}|4\rangle = \langle u_m|\varphi\rangle. \rightarrow \text{右边: } |\varphi\rangle = \sum_n b_n|u_n\rangle.$$

$$\therefore \langle u_m|\varphi\rangle = b_m$$

$$\rightarrow \text{左边: } \langle u_m|\hat{Q}|4\rangle = \sum_n \underbrace{\langle u_m|\hat{Q}|u_n\rangle}_{\downarrow} \underbrace{\langle u_n|\varphi\rangle}_{\downarrow} \rightarrow \begin{cases} |\varphi\rangle = \sum_n a_n|u_n\rangle \\ a_n = \langle u_n|\varphi\rangle \end{cases}$$

$$= \sum_n Q_{mn} a_n$$

$$\therefore \sum_n Q_{mn} a_n = b_m \rightarrow \text{即: } \begin{pmatrix} Q_{11} & Q_{12} & \dots \\ Q_{21} & Q_{22} & \dots \\ \vdots & \vdots & \ddots \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix} \quad \star \quad \underline{Q_{mn} = \langle u_m|\hat{Q}|u_n\rangle.}$$

力学量在自身表象下的矩阵元: $F_{mn} = \langle u_m|\hat{F}|u_n\rangle$

$$= f_n \langle u_m|u_n\rangle \Rightarrow \hat{F} = \begin{pmatrix} f_1 & & \\ & f_2 & \\ & & \ddots \\ & & & f_n \end{pmatrix}$$

$$= f_n \delta_{mn}$$

(二). 坐标表象.

由量子力学基本原理给出:
$$\begin{cases} \hat{r} = \vec{r} \\ \hat{p} = -i\hbar \vec{\nabla} \end{cases}$$

11. 坐标算符.

-维: $\hat{x}|x'\rangle = x'|x'\rangle$

$$|x'\rangle = \delta(x - x')$$

$$\langle x'|x''\rangle = \delta(x' - x'')$$

$$\therefore |4\rangle = \int C_x |x'\rangle dx'$$

$\downarrow$

$$C_x = \langle x'|4\rangle = \psi(x', t). \Rightarrow \text{玻恩的统计解释.}$$

12. 动量算符.

$$\hat{p}_x = -i\hbar \frac{\partial}{\partial x}$$

$$\hat{p}_x|\varphi_{p_x}\rangle = p_x|\varphi_{p_x}\rangle$$

$$\Rightarrow |\varphi_{p_x}\rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{\frac{i}{\hbar} p_x x}$$

$$\langle \varphi_{p_x}|\varphi_{p_x'}\rangle = \delta(p_x - p_x')$$

不应是 $\delta(p_x' - p_x)$ 吗?

如果无所谓, 那谁是变量呢?

(三). 动量表象:

(1). 动量算符:

$$\langle p'_x | p_x \rangle = \int \langle p'_x | x \rangle \langle x | p_x \rangle dx \quad \rightarrow \text{若 } p_x \text{ 不是连续谱呢? } \langle p'_x | p_x \rangle = ?$$

$$= \delta(p'_x - p_x)$$

$$\therefore \text{矩阵元: } (\hat{p}_x)_{p'_x p_x} = \langle p'_x | \hat{p}_x | p_x \rangle$$

$$\text{相当于行、列指标} = p'_x \langle p'_x | p_x \rangle$$

$$= p'_x \delta(p'_x - p_x) = p_x \delta(p'_x - p_x) \Rightarrow \text{对角阵.}$$

$\hat{p}_x$ 的“函数”表示:

$$\text{如 } | \psi \rangle, \quad \hat{p}_x | \psi \rangle = | \psi \rangle$$

$$\text{在动量表象下展开: } | \psi \rangle = \int b(p_x) | p_x \rangle dp_x$$

$$= \int b(p_x, t) | p_x \rangle dp_x$$

$$b(p_x, t) = \langle p_x | \psi \rangle$$

$$\text{同理: } | \psi \rangle = \int a(p_x, t) | p_x \rangle dp_x, \quad a(p_x, t) = \langle p_x | \psi \rangle$$

$$\therefore \langle p_x | \hat{p}_x | \psi \rangle = \langle p_x | \psi \rangle$$

$$\text{左边: } \langle p_x | \hat{p}_x | \psi \rangle = \int \langle p_x | \hat{p}_x | p'_x \rangle \langle p'_x | \psi \rangle dp'_x$$

$$= \int p'_x \delta(p_x - p'_x) a(p'_x, t) dp'_x$$

$$= p_x a(p_x, t)$$

$$\text{右边: } \langle p_x | \psi \rangle = b(p_x, t)$$

$$\therefore p_x a(p_x, t) = b(p_x, t) \quad \text{对比: } \hat{p}_x | \psi \rangle = | \psi \rangle$$

$$\Rightarrow \hat{p}_x = p_x$$

(2). 坐标算符:

$$\text{矩阵元: } (\hat{x})_{p'_x p_x} = \langle p'_x | \hat{x} | p_x \rangle$$

$$= \iint \langle p'_x | x \rangle \langle x | \hat{x} | x' \rangle \langle x' | p_x \rangle dx dx'$$

$$= \iint \frac{1}{\sqrt{2\pi\hbar}} e^{-\frac{i}{\hbar} p'_x x} x \delta(x - x') e^{\frac{i}{\hbar} p_x x'} dx dx'$$

$$= \frac{1}{\sqrt{2\pi\hbar}} \int x e^{\frac{i}{\hbar} (p'_x - p_x) x} dx$$

$$= (-i\hbar) \frac{\partial}{\partial p_x} \int \psi_{p'_x}^*(x) \psi_{p_x}(x) dx$$

$$= \underline{-i\hbar \frac{\partial}{\partial p_x} \delta(p'_x - p_x)} = \underline{i\hbar \frac{\partial}{\partial p_x} \delta(p'_x - p_x)}$$

$\hat{x}$ 的函数表示:

$$\text{如 } | \psi \rangle, \quad \hat{x} | \psi \rangle = | \psi \rangle$$

$$\therefore \langle p_x | \hat{x} | \psi \rangle = \langle p_x | \psi \rangle$$

同理可得:

$$+i\hbar \frac{\partial}{\partial p_x} a(p_x, t) = b(p_x, t)$$

$$\therefore \hat{x} = +i\hbar \frac{\partial}{\partial p_x}$$

$$\therefore \text{动量表象下: } \begin{cases} \hat{x} = i\hbar \frac{\partial}{\partial p_x} \\ \hat{p}_x = p_x \end{cases} \rightarrow \begin{cases} \hat{r} = i\hbar \vec{\nabla}_p \\ \hat{p} = \vec{p} \end{cases}$$

(四) $\hat{F}$ 表象下的量子力学公式

(1) 本征值方程:

$$\hat{Q}|\psi\rangle = \lambda|\psi\rangle$$

$$\langle u_m | \hat{Q} |\psi\rangle = \lambda \langle u_m | \psi \rangle$$

$$\langle u_m | \hat{Q} | u_n \rangle \langle u_n | \psi \rangle = \lambda \langle u_m | \psi \rangle$$

$$\sum_n \hat{Q}_{mn} a_n = \lambda a_m$$

本征值方程在某表象下即为矩阵的本征值方程.

$$\Rightarrow \begin{pmatrix} Q_{11} & Q_{12} & \dots \\ Q_{21} & Q_{22} & \dots \\ \vdots & \vdots & \ddots \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \\ \vdots \end{pmatrix} = \lambda \begin{pmatrix} a_1 \\ a_2 \\ \vdots \end{pmatrix}$$

(2) 平均值.

$$\bar{F} = \langle \psi | \hat{F} | \psi \rangle, |\psi\rangle = \sum_n c_n |u_n\rangle, c_n = \langle u_n | \psi \rangle$$

$$\therefore \bar{F} = \sum_n |c_n|^2 f_n$$

$$= \sum_n c_n^* c_n f_n$$

$$= \sum_n \langle \psi | u_n \rangle \langle u_n | \psi \rangle f_n$$

$$= \sum_n \langle \psi | \hat{F} | u_n \rangle \langle u_n | \psi \rangle$$

$$= \langle \psi | \hat{F} | \psi \rangle$$

求平均值一定得在自身表象下用自己的本征矢展开, 或者说可以在别的表象下但依然得用自身本征矢展开!

(3) 薛定谔方程.

$$i\hbar \frac{\partial}{\partial t} |\psi\rangle = \hat{H} |\psi\rangle$$

$$\text{在 } \hat{F} \text{ 表象下: } i\hbar \frac{\partial}{\partial t} \langle u_m | \psi \rangle = \hat{H}_{m\mu} \langle u_\mu | \psi \rangle$$

$$= \sum_n \hat{H}_{mn} \langle u_n | \psi \rangle$$

$$i\hbar \frac{\partial}{\partial t} a_m = \sum_n H_{mn} a_n$$

也即: $i\hbar \frac{\partial}{\partial t} \begin{pmatrix} a_1 \\ a_2 \\ \vdots \end{pmatrix} = \begin{pmatrix} H_{11} & H_{12} & \dots \\ H_{21} & H_{22} & \dots \\ \dots & \dots & \ddots \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \\ \vdots \end{pmatrix}$

在例 2 中若要求 t 时刻的 $\hat{A}$ 的取值概率咋办?

三. 表象变换.

$$\hat{A}^\dagger = \hat{A}, \hat{A}|a_n\rangle = a_n|a_n\rangle$$

$$\hat{B}^\dagger = \hat{B}, \hat{B}|b_n\rangle = b_n|b_n\rangle$$

(一) 态矢量的变换.

$$\hat{A} \text{ 表象下: } |\psi\rangle = \sum_n c_n |a_n\rangle, c_n = \langle a_n | \psi \rangle$$

$$\therefore |\psi\rangle = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \end{pmatrix}$$

$$\hat{B} \text{ 表象下: } |\psi\rangle = \sum_n c'_n |b_n\rangle, c'_n = \langle b_n | \psi \rangle$$

$$\therefore |\psi\rangle = \begin{pmatrix} c'_1 \\ c'_2 \\ \vdots \end{pmatrix}$$

$$\begin{aligned}
 \therefore C_m &= \langle a_m | \psi \rangle \\
 &= \sum_n \langle a_m | b_n \rangle \langle b_n | \psi \rangle \\
 &= \sum_n S_{mn} C_n'
 \end{aligned}$$

$$\therefore S_{mn} = \langle a_m | b_n \rangle$$

$$\begin{pmatrix} C_1 \\ C_2 \\ \vdots \end{pmatrix} = \begin{pmatrix} S_{11} & S_{12} & \cdots \\ S_{21} & S_{22} & \cdots \\ \cdots & \cdots & \cdots \end{pmatrix} \begin{pmatrix} C_1' \\ C_2' \\ \vdots \end{pmatrix}$$

$$\Rightarrow \begin{cases} |\psi\rangle^A = \hat{S} |\psi\rangle^B \\ S_{mn} = \langle a_m | b_n \rangle \end{cases}$$

* $\hat{S}$ 的一些性质:

$$\textcircled{1} \quad \tilde{S}_{mn} = S_{nm} = \langle a_n | b_m \rangle$$

$$\textcircled{2} \quad \tilde{S}_{mn}^* = (\hat{S}^\dagger)_{mn} = \langle a_n | b_m \rangle^* = \langle b_m | a_n \rangle$$

$$\begin{aligned}
 \textcircled{3} \quad (\hat{S} \hat{S}^\dagger)_{mn} &= \sum_\alpha \hat{S}_{m\alpha} (\hat{S}^\dagger)_{\alpha n} \\
 &= \sum_\alpha \langle a_m | b_\alpha \rangle \langle b_\alpha | a_n \rangle \\
 &= \langle a_m | a_n \rangle \\
 &= \delta_{mn}
 \end{aligned}$$

$\therefore \hat{S} \hat{S}^\dagger = I \Rightarrow \hat{S}$ 是么正矩阵. $\Rightarrow$ 表象变换即么正变换. $\hat{S}$ 矩阵即把 B 表象下的态矢用 A 表象的基矢表示.

$$\textcircled{4} \quad \hat{S} \text{ 的第 } n \text{ 列: } \begin{pmatrix} \langle a_1 | b_n \rangle \\ \langle a_2 | b_n \rangle \\ \vdots \\ \langle a_m | b_n \rangle \end{pmatrix} \Rightarrow |b_n\rangle \text{ 在 } A \text{ 表象下的表示}$$

(二) 力学量的变换.

$$\hat{F}^\dagger = \hat{F} \quad A \text{ 表象下: } \langle a_m | \hat{F} | a_n \rangle = F_{mn}^A$$

$$B : \langle b_m | \hat{F} | b_n \rangle = F_{mn}^B$$

$$\begin{aligned}
 \therefore F_{mn}^A &= \langle a_m | \hat{F} | a_n \rangle \\
 &= \sum_{\alpha\beta} \langle a_m | b_\alpha \rangle \langle b_\alpha | \hat{F} | b_\beta \rangle \langle b_\beta | a_n \rangle \\
 &= \sum_{\alpha\beta} S_{m\alpha} F_{\alpha\beta}^B S_{\beta n}^\dagger
 \end{aligned}$$

$$\begin{aligned}
 \therefore \hat{F}^A &= \hat{S} \hat{F}^B \hat{S}^\dagger \\
 |\psi\rangle^A &= \hat{S} |\psi\rangle^B, \quad S_{mn} = \langle a_m | b_n \rangle
 \end{aligned}$$

基矢的变换:

$$|a_m\rangle = \sum_n d_n |b_n\rangle, \quad d_n = \langle a_m | b_n \rangle = (\hat{S}^\dagger)_{nm}$$

$$\begin{aligned} \therefore |a_m\rangle &= \sum_n (\hat{S}^\dagger)_{nm} |b_n\rangle \\ &= \sum_n (\hat{S}^\dagger)_{mn} |b_n\rangle \\ &= \sum_n (S^*)_{mn} |b_n\rangle \end{aligned}$$

四). 么正变换的性质:

$$\begin{cases} \hat{F}^A = \hat{S} \hat{F}^B \hat{S}^\dagger \\ |a\rangle^A = \hat{S} |a\rangle^B, \quad S_{mn} = \langle a_m | b_n \rangle \end{cases}$$

(1). 不改变算符的厄米性.

(2). 不改变两个矢量的标量积. $\rightarrow$ 也即归一化, 正交性不受影响

(3). 不改变力学量的本征值.

(4). 不改变矩阵的迹.

(5). 不改变量子力学中矢量方程的形式

$\Rightarrow$ 么正变换下量子力学不变.

四. 粒子数表象 (线性谐振子)

1-2 二个新算符:

$$\hat{H} = \frac{\hat{p}^2}{2\mu} + \frac{1}{2}\mu\omega^2 \hat{x}^2, \quad (\hat{p} = \hat{p}_x)$$

$$\begin{aligned} \therefore \text{无量纲化: } \frac{\hat{H}}{\hbar\omega} &= \frac{\hat{p}^2}{2\hbar\mu\omega} + \frac{\mu\omega}{2\hbar} \hat{x}^2 \\ &= \frac{\mu\omega}{2\hbar} (\hat{x}^2 + \frac{1}{\mu\omega} \hat{p}^2) \end{aligned}$$

☆

$$\text{记: } \hat{a} = \sqrt{\frac{\mu\omega}{2\hbar}} (\hat{x} + \frac{i}{\mu\omega} \hat{p})$$

$$\Rightarrow \hat{a} = \frac{\alpha}{\sqrt{2}} (\hat{x} + \frac{i}{\mu\omega} \hat{p}), \quad \alpha = \sqrt{\frac{\mu\omega}{\hbar}}$$

$$\hat{a}^\dagger = \frac{\alpha}{\sqrt{2}} (\hat{x} - \frac{i}{\mu\omega} \hat{p}).$$

$$\therefore [\hat{a}, \hat{a}^\dagger] = 1.$$

$$\therefore \begin{cases} \hat{x} = \frac{1}{\sqrt{2}\alpha} (\hat{a}^\dagger + \hat{a}) \\ \hat{p} = \frac{i\hbar\alpha}{\sqrt{2}} (\hat{a}^\dagger - \hat{a}) \end{cases}$$

$$\begin{aligned} \therefore \hat{H} &= \frac{1}{2\mu} \cdot \frac{\hbar^2 \alpha^2}{2} (\hat{a}^\dagger - \hat{a})^2 + \frac{1}{2}\mu\omega^2 \cdot \frac{1}{2\alpha^2} (\hat{a}^\dagger + \hat{a})^2 \\ &= \frac{\hbar\omega}{2} (\hat{a}^\dagger \hat{a} + \hat{a} \hat{a}^\dagger) \end{aligned}$$

$$= (\hat{a}^\dagger \hat{a} + \frac{1}{2}) \hbar\omega$$

$$\text{记: } \hat{N} = \hat{a}^\dagger \hat{a}, \quad \text{有: } \hat{N}^\dagger = \hat{N}$$

$$\therefore \hat{H} = (\hat{N} + \frac{1}{2}) \hbar\omega$$

$$\therefore \hat{N} = \hat{a}^\dagger \hat{a}$$

$$\therefore \text{在 } |n\rangle \text{ 下: } \langle n | \hat{N} | n \rangle = n$$

$$\therefore \hat{H} \geq \frac{1}{2} \hbar\omega$$

(二). 粒子数表象.

$$\hat{N}|\lambda\rangle = \lambda|\lambda\rangle$$

$\begin{cases} x \rightarrow \infty \text{ 时 } \hat{H} \text{ 有限: 束缚态,} \\ \text{-维谐振子: 非简并.} \end{cases}$

$$[\hat{N}, \hat{a}] = -\hat{a}$$

$$\therefore \hat{N}\hat{a} - \hat{a}\hat{N} = -\hat{a}$$

$$\therefore \hat{N}\hat{a} = \hat{a}\hat{N} - \hat{a}$$

$$\begin{aligned} \therefore \hat{N}\hat{a}|\lambda\rangle &= \hat{a}\hat{N}|\lambda\rangle - \hat{a}|\lambda\rangle \\ &= (\lambda-1)\hat{a}|\lambda\rangle, \dots \textcircled{1} \end{aligned}$$

可推出两点:

①. $\hat{a}|\lambda\rangle = 0$ (特殊情况成立)

②. $\hat{a}|\lambda\rangle = C|\lambda-1\rangle$ (一般情况成立)

对: ②. $\lambda-1$ 和 λ 是 $\hat{N}$ 的本征值, 则 $\lambda-2, \lambda-3 \dots$ 也是 $\hat{N}$ 的本征值.

同理: $[\hat{N}, \hat{a}^+] = \hat{a}^+$

$$\begin{aligned} \therefore \hat{N}\hat{a}^+|\lambda\rangle &= \hat{a}^+|\lambda\rangle + \hat{a}^+\hat{N}|\lambda\rangle \\ &= (\lambda+1)\hat{a}^+|\lambda\rangle, \dots \textcircled{2} \end{aligned}$$

$$\therefore \hat{a}^+|\lambda\rangle = C'|\lambda+1\rangle$$

$\therefore \lambda+1, \lambda+2 \dots$ 也是 $\hat{N}$ 的本征值.

$$\therefore \lambda \Rightarrow \lambda \pm 1, \lambda \pm 2, \dots$$

又: $\forall |4\rangle: \hat{N} = \langle 4|\hat{N}|4\rangle \geq 0$

$\therefore$ 取 $|4\rangle$ 为 $|\lambda\rangle$, 有: $\lambda \geq 0$

$\therefore \lambda$ 有最小值 λ_0 , 对应本征矢为 $|\lambda_0\rangle$

由: ① 知:

$$\hat{N}\hat{a}|\lambda_0\rangle = (\lambda_0-1)\hat{a}|\lambda_0\rangle$$

λ_0-1 不存在. $\therefore \hat{a}|\lambda_0\rangle = 0$

$$\therefore \hat{a}^+\hat{a}|\lambda_0\rangle = 0$$

即 $\hat{N}|\lambda_0\rangle = 0 = \lambda_0|\lambda_0\rangle \Rightarrow \lambda_0 = 0$

$\therefore \hat{N}$ 的本征值: $\lambda_0, \lambda_0+1, \dots \Rightarrow 0, 1, 2, \dots = n$ (自然数)

$\therefore \hat{H}$ 的本征值为 $(n \pm \frac{1}{2})\hbar\omega$. $\rightarrow$ 非常简并自然地给出了 - 维线性谐振子的能级.

定义: $\hat{N}$ 为“^{算符}粒子数表象”, $\{|n\rangle\}$ 为粒子数表象. $\rightarrow$ “声子” $\Leftrightarrow$ 简谐振动

$$\therefore \begin{cases} \hat{N}|n\rangle = n|n\rangle \end{cases}$$

$$\begin{cases} \hat{H}|n\rangle = (n \pm \frac{1}{2})\hbar\omega|n\rangle \rightarrow \text{定薛,} \end{cases}$$

(三). 产生算符与湮灭算符

$$\begin{cases} \hat{a}|n\rangle = c|n-1\rangle & \Rightarrow \hat{a} \text{ 为粒子的湮灭算符} \\ \hat{a}^+|n\rangle = c'|n+1\rangle & \Rightarrow \hat{a}^+ \text{ 产生算符} \end{cases}$$

对 c, c' 讨论:

在左矢空间: $\begin{cases} \langle n|\hat{a}^+ = \langle n-1|c^* & \hat{a}^+: \text{湮灭} \\ \langle n|\hat{a} = \langle n+1|c'^* & \hat{a}: \text{产生} \end{cases}$

$$\begin{aligned} \therefore \langle n|\hat{a}^+\hat{a}|n\rangle &= \langle n-1|c^*c|n-1\rangle = |c|^2 \\ \text{"} \\ \langle n|\hat{a}\hat{a}^+|n\rangle &= n\langle n|n\rangle = n \end{aligned} \quad \left. \vphantom{\begin{aligned} \therefore \langle n|\hat{a}^+\hat{a}|n\rangle = \dots \end{aligned}} \right\} n = |c|^2 \Rightarrow c = \sqrt{n}$$

同理可得: $c' = \sqrt{n+1}$

$$\begin{cases} \hat{a}|n\rangle = \sqrt{n}|n-1\rangle \\ \hat{a}^+|n\rangle = \sqrt{n+1}|n+1\rangle \end{cases}$$

(四). 坐标表象下的本征矢量

$$\hat{a}|\lambda\rangle = 0 \quad \text{即: } \hat{a}|0\rangle = 0$$

在坐标表象下: $\hat{a} = \frac{\alpha}{\sqrt{2}}(\hat{x} + \frac{\hbar}{i m \omega} \frac{d}{dx})$

设: $\varphi_0(x) = \langle x|0\rangle$

$$\therefore \frac{\alpha}{\sqrt{2}}(\hat{x} + \frac{\hbar}{i m \omega} \frac{d}{dx})\varphi_0(x) = 0$$

$$\Rightarrow \varphi_0 = A e^{-\frac{i}{2}\alpha^2 x^2}$$

归一化得: $A = \sqrt{\frac{\alpha}{\hbar}} = \left(\frac{m\omega}{\hbar}\right)^{\frac{1}{4}}$

而 $\hat{a}^+|0\rangle = |1\rangle$

设 $\varphi_1(x) = \langle x|1\rangle$

$$\therefore \varphi_1(x) = \frac{\alpha}{\sqrt{2}}(\hat{x} - \frac{\hbar}{i m \omega} \frac{d}{dx})\varphi_0(x)$$

由 $\hat{a}^+|n\rangle = \sqrt{n+1}|n+1\rangle$

$$\therefore |n\rangle = \frac{1}{\sqrt{n!}}(\hat{a}^+)^n|0\rangle$$

应用:

$$\begin{aligned} (1) \hat{x}|n\rangle &= \frac{1}{\sqrt{2}\alpha}(\hat{a}^+ + \hat{a})|n\rangle \\ &= \frac{1}{\sqrt{2}\alpha}(\sqrt{n+1}|n+1\rangle + \sqrt{n}|n-1\rangle) \end{aligned}$$

$$\begin{aligned} (2) \hat{p}|n\rangle &= \frac{i\hbar\alpha}{\sqrt{2}}(\sqrt{n+1}\hat{a}^+ - \hat{a})|n\rangle \\ &= \frac{i\hbar\alpha}{\sqrt{2}}(\sqrt{n+1}|n+1\rangle - \sqrt{n}|n-1\rangle) \end{aligned}$$

(3) 表象下的矩阵元:

$$\langle m|\hat{x}|n\rangle = \frac{\alpha}{\sqrt{2}\alpha}(\sqrt{n+1}\delta_{m,n+1} + \sqrt{n}\delta_{m,n-1})$$

第九章 自旋和角动量基础

一、角动量算符

(1) $\vec{L} = \vec{r} \times \vec{p} \Rightarrow$ 轨道角动量: $\vec{L} = \vec{r} \times \vec{p}$

$[\hat{L}_\alpha, \hat{L}_\beta] = i\hbar \epsilon_{\alpha\beta\gamma} \hat{L}_\gamma$

(一) 角动量算符的定义

☆ 1. 定义: 一个矢量算符, 它的三个分量满足: $[\hat{J}_\alpha, \hat{J}_\beta] = i\hbar \epsilon_{\alpha\beta\gamma} \hat{J}_\gamma$, 则称 $\hat{J}$ 为角动量.

(2) $\hat{J}^2 = \hat{J}_x^2 + \hat{J}_y^2 + \hat{J}_z^2 \rightarrow \hat{J}^2$ 与 $\hat{J}_z$ 可同时取确定值, 但角动量分量不能同时确定 $\Rightarrow$ means $\hat{J}$ 方向不定
 $\Rightarrow [\hat{J}^2, \hat{J}_\alpha] = 0$

(二) $\hat{J}^2$ 与 $\hat{J}_z$ 的共同本征态

(1) 设共同本征态为: $|\lambda m\rangle$

$\hat{J}^2 |\lambda m\rangle = \lambda \hbar^2 |\lambda m\rangle$

$\hat{J}_z |\lambda m\rangle = m \hbar |\lambda m\rangle$

核心问题: 求解: $\hat{J}_x |\lambda m\rangle = ?$ $\hat{J}_y |\lambda m\rangle = ?$

$\Downarrow$

☆ 12. $\langle \lambda m | \hat{J}_x^2 | \lambda m \rangle = \langle \lambda m | \hat{J}_x^+ \hat{J}_x | \lambda m \rangle \geq 0 \rightarrow$ 因为力学量的本征值是实数, 平方后必然非负

同理: $\langle \lambda m | \hat{J}_y^2 | \lambda m \rangle \geq 0$

$\therefore \langle \lambda m | \hat{J}^2 | \lambda m \rangle = \langle \lambda m | \hat{J}_x^2 + \hat{J}_y^2 + \hat{J}_z^2 | \lambda m \rangle \geq \langle \lambda m | \hat{J}_z^2 | \lambda m \rangle$

$\Rightarrow \lambda \hbar^2 \geq m^2 \hbar^2$

$\therefore \lambda \geq m^2 \rightarrow$ means: ① $\lambda \geq 0$

② m 有上下限: $-\sqrt{\lambda} \leq m \leq \sqrt{\lambda}$

定义: $\hat{J}_\pm = \hat{J}_x \pm i\hat{J}_y$

性质: ① $(\hat{J}_\pm)^\dagger = \hat{J}_\mp$

☆ ② $\hat{J}_+ \hat{J}_- = \hat{J}^2 - \hat{J}_z^2 + \hbar \hat{J}_z$

$\hat{J}_- \hat{J}_+ = \hat{J}^2 - \hat{J}_z^2 - \hbar \hat{J}_z$

$\therefore [\hat{J}_+, \hat{J}_-] = 2\hbar \hat{J}_z$

③ $\begin{cases} \hat{J}_x = \frac{1}{2}(\hat{J}_+ + \hat{J}_-) \\ \hat{J}_y = \frac{1}{2i}(\hat{J}_+ - \hat{J}_-) \end{cases}$

☆ ④ $[\hat{J}_z, \hat{J}_\pm] = \pm \hbar \hat{J}_\pm$

$\Downarrow$

$\hat{J}_z \hat{J}_\pm |\lambda m\rangle = (\pm \hbar \hat{J}_\pm + \hat{J}_z \hat{J}_\pm) |\lambda m\rangle$
 $= (m \pm 1) \hbar \hat{J}_\pm |\lambda m\rangle$

$\therefore \hat{J}_\pm |\lambda m\rangle$ 是 $\hat{J}_z$ 本征值为 $(m \pm 1)\hbar$ 的本征矢.

又: $[\hat{J}^2, \hat{J}_\pm] = 0$

$\therefore \hat{J}^2 \hat{J}_\pm |\lambda m\rangle = \hat{J}_\pm \hat{J}^2 |\lambda m\rangle$
 $= \lambda \hbar^2 \hat{J}_\pm |\lambda m\rangle$

$\therefore \hat{J}_\pm |\lambda m\rangle$ 也是 $\hat{J}^2$ 的本征值为 $\lambda \hbar^2$ 的本征矢.

综上: $\hat{J}_\pm |\lambda m\rangle = C_{\pm} |\lambda, m \pm 1\rangle \Rightarrow \hat{J}_\pm$ 为 $\hat{J}$ 的上升/下降算符

$C_{\pm}$ 如何求解

由 m 有上下限, 设上限为 A , 下限为 B

$$\therefore \begin{cases} \hat{J}_+ |\lambda A\rangle = 0 & \dots \textcircled{a} \\ \hat{J}_- |\lambda B\rangle = 0 & \dots \textcircled{b} \end{cases}$$

对 $\textcircled{a}$: $\hat{J}_- \hat{J}_+ |\lambda A\rangle = 0$

$$\therefore (\hat{J}^2 - \hat{J}_z^2 - \hbar \hat{J}_z) |\lambda A\rangle = 0 \Rightarrow \lambda = A(A+1)$$

同理对 $\textcircled{b}$ 有: $\hat{J}_+ \hat{J}_- |\lambda B\rangle = 0 \Rightarrow \lambda = B(B-1)$

$$\therefore A(A+1) = B(B-1)$$

$$\Rightarrow A = B-1 \quad (\times)$$

$$A = -B \quad (\checkmark)$$

$$\therefore \text{令: } A = -B = j \Rightarrow \lambda = A(A+1) = j(j+1)$$

$$-j \leq m \leq j \Rightarrow m = j, j-1, \dots, -j \quad (2j+1 \text{ 个取值})$$

$$\therefore |\lambda m\rangle \rightarrow |jm\rangle$$

$$\therefore \begin{cases} \hat{J}^2 |jm\rangle = j(j+1)\hbar^2 |jm\rangle \\ \hat{J}_z |jm\rangle = m\hbar |jm\rangle \end{cases}$$

j 取什么值?

$$\therefore A - B = n \quad (\text{自然数})$$

$$\therefore 2j = n$$

$$\Rightarrow j = \frac{1}{2}n = 0, \frac{1}{2}, 1, \frac{3}{2}, \dots$$

$$m = j, j-1, \dots, -j$$

角量子数 (不是说一定这样取值, 而是只能取这些值)

磁量子数

$$\therefore \hat{J}_{\pm} |jm\rangle = C_{\pm} |j, m\pm 1\rangle$$

$$\langle jm | \hat{J}_{\mp} = \langle j, m\pm 1 | C_{\pm}^*$$

$$\therefore \langle jm | \hat{J}_{\mp} \hat{J}_{\pm} |jm\rangle = |C_{\pm}|^2 \quad (C_{\pm} \text{ 为实数})$$

$$\therefore C_{\pm}^2 = \langle jm | \hat{J}^2 - \hat{J}_z^2 \mp \hbar \hat{J}_z |jm\rangle$$

$$= \hbar^2 [j(j+1) - m(m\pm 1)]$$

$$\Rightarrow C_{\pm} = \sqrt{j(j+1) - m(m\pm 1)} \hbar$$

$$\therefore \hat{J}_{\pm} |jm\rangle = \sqrt{j(j+1) - m(m\pm 1)} \hbar |j, m\pm 1\rangle$$

$$\text{那么 } \hat{J}_x |jm\rangle = ? \quad \hat{J}_y |jm\rangle = ?$$

(三). $\hat{J}_1$ 和 $\hat{J}_2$ 共同表象下的角动量算符的矩阵形式 (四维)

$$(\hat{J}_1)_{j'm'jm} = \langle j'm' | \hat{J}_1 | j'm \rangle$$

$$= j(j+1)\hbar^2 \delta_{j'm'jm} \delta_{jj} \delta_{mm}$$

$$(\hat{J}_2)_{j'm'jm} = \langle j'm' | \hat{J}_2 | j'm \rangle$$

$$= m\hbar \delta_{jj'} \delta_{mm'}$$

$$(\hat{J}_x)_{j'm'jm} = \langle j'm' | \frac{1}{2}(\hat{J}_+ + \hat{J}_-) | j'm \rangle$$

$$\Leftrightarrow (\hat{J}_\pm)_{j'm'jm} = \hbar \sqrt{j(j+1) - m(m\pm 1)} \delta_{jj'} \delta_{m'm\pm 1}$$

$$(\hat{J}_y)_{j'm'jm} = \langle j'm' | \frac{1}{2i}(\hat{J}_+ - \hat{J}_-) | j'm \rangle$$

二. 电子的自旋.

1. 实验基础

2. 电子的自旋

电子的自旋假定: { ①. 电子具有自旋角动量 $\vec{S}$, $\vec{S}$ 在任何方向的投影只可能有两个取值: $\pm \frac{\hbar}{2}$.
②. 电子具有自旋磁矩:

$$\vec{\mu} = -\frac{e}{2m} \vec{S} \Rightarrow \mu_z = -\frac{e}{2m} S_z = \pm \frac{e\hbar}{2m} = \pm \mu_B \text{ (玻尔磁子).}$$

此即电子自旋的定义, 是电子的内禀属性.

(三). 自旋算符的形式:

$$\hat{S}_z = \pm \frac{\hbar}{2} \Leftrightarrow \hat{J}_z = m\hbar \quad (m = j, j-1, \dots, -j)$$

$\therefore m_s = \pm \frac{1}{2}$ (自旋磁量子数) $\Rightarrow j = \frac{1}{2}$. 用 S 表示即: $S = \frac{1}{2}$ (自旋量子数)

$\therefore |jm\rangle \rightarrow |S m_s\rangle$: $\vec{S}$ 和 $\hat{S}_z$ 的共同本征态. (对电子可简写为 $|\frac{1}{2} m_s\rangle$)

↓

$$|S, m_s\rangle \Rightarrow |\frac{1}{2} m_s\rangle \Rightarrow \begin{cases} |\frac{1}{2} \frac{1}{2}\rangle \\ |\frac{1}{2} -\frac{1}{2}\rangle \end{cases} \Rightarrow \begin{cases} |+\rangle \\ |-\rangle \end{cases}$$

$$m. \hat{S}^2 |+\rangle = \frac{1}{2} \times \frac{3}{2} \hbar^2 |+\rangle = \frac{3}{4} \hbar^2 |+\rangle$$

$$\hat{S}_z |+\rangle = \frac{\hbar}{2} |+\rangle$$

$$\hat{S}_z |-\rangle = -\frac{\hbar}{2} |-\rangle$$

$$\begin{cases} |+\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ |-\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \end{cases}$$

$\therefore$ 在 $\hat{S}^2$ 和 $\hat{S}_z$ 的共同表象下:

$$\hat{S}^2 = \frac{3}{4} \hbar^2 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \frac{3}{4} \hbar^2 I$$

$$\hat{S}_z = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$\hat{S}_x = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$\hat{S}_y = \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

12) 波函数 $14\rangle$

“自旋是第四自由度, 描述 $14\rangle$ 除了坐标现在要引入自旋”

∴ 在坐标表象下: $14\rangle = \psi(\vec{r}, S_z, t) = \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix} \quad \begin{cases} \psi_1 = \langle +14 \rangle \\ \psi_2 = \langle -14 \rangle \end{cases}$

$$= \psi_1 |+\rangle + \psi_2 |-\rangle$$

∴ $\int |\psi|^2 d\tau = 1$ 表示全空间自旋为 $\pm$ 的概率.

②. 力学量:

$$\hat{G} = \begin{pmatrix} \hat{G}_{11} & \hat{G}_{12} \\ \hat{G}_{21} & \hat{G}_{22} \end{pmatrix} \rightarrow \begin{cases} \hat{G}_{11} = \hat{G}_{11}(\vec{r}, \hat{p}, t) = \langle +1 | \hat{G} | + \rangle \\ \hat{G}_{21} = \langle -1 | \hat{G} | + \rangle \\ \hat{G}_{12} = \langle +1 | \hat{G} | - \rangle \\ \hat{G}_{22} = \langle -1 | \hat{G} | - \rangle \end{cases}$$

⇒ 其实就是力学量在某表象下的矩阵表示. m_s 只有 2 个取值, 因此为 2×2 矩阵

∴ $\langle \hat{G} \rangle = \langle 4 | \hat{G} | 4 \rangle = \int (\psi_1^* \psi_2^*) \begin{pmatrix} \hat{G}_{11} & \hat{G}_{12} \\ \hat{G}_{21} & \hat{G}_{22} \end{pmatrix} \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix} d\tau$

13) 泡利算符 (为了无量纲化)

①. 定义: $\hat{\sigma} = \frac{2}{\hbar} \hat{S}$, $\hat{\sigma}$: Pauli 算符.

∴ 在 $\hat{\sigma}_z$ 和 $\hat{\sigma}_x$ 的共同表象下

$$\begin{cases} \hat{\sigma}_z = 3 \\ \hat{\sigma}_x = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \\ \hat{\sigma}_y = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \end{cases}$$

②. 性质: (i). 代数关系:

- 1°. 都是厄米的
- 2°. $\hat{\sigma}_x^2 = \hat{\sigma}_y^2 = \hat{\sigma}_z^2 = 1$
- 3°. $\text{tr}(\hat{\sigma}_a) = 0$

[L_x, L_y] 类比 →

- 4°. $[\hat{\sigma}_a, \hat{\sigma}_b] = 2i \epsilon_{abc} \hat{\sigma}_c$
- 5°. 反对易关系: $\{\hat{\sigma}_a, \hat{\sigma}_b\} = 2\delta_{ab}$

- 6°. $\hat{\sigma}_x \hat{\sigma}_y = i\hat{\sigma}_z$, $\hat{\sigma}_y \hat{\sigma}_x = -i\hat{\sigma}_z$
- 7°. $\hat{\sigma}_x \hat{\sigma}_y \hat{\sigma}_z = i$

可证出: 与两个泡利矩阵反对易的矩阵必与第三个泡利矩阵对易

- (ii). 与 $\hat{\sigma}_x, \hat{\sigma}_y, \hat{\sigma}_z$ 均反对易的 2×2 矩阵必为 0 矩阵.
- (iii). 与三个泡利矩阵中的两个对易, 必与第三个对易
- (iv). 任何 2×2 矩阵均可写成 $\hat{\sigma}_x, \hat{\sigma}_y, \hat{\sigma}_z, \hat{I}$ 的线性组合
- (v). 与三个泡利矩阵均对易的 2×2 矩阵一定是常数矩阵
- (vi). 若 $[\hat{\sigma}, \hat{A}] = 0$, $[\hat{\sigma}, \hat{B}] = 0$, 则有:

$$(\hat{\sigma} \cdot \hat{A})(\hat{\sigma} \cdot \hat{B}) = \hat{A} \cdot \hat{B} + i\hat{\sigma} \cdot (\hat{A} \times \hat{B})$$

(第二道例题改前)

三. 角动量的耦合

(一). 两个角动量的和

有两个相互独立的角动量 $\hat{J}_1, \hat{J}_2$, means $[\hat{J}_1, \hat{J}_2] = 0$ (矢量算符对易 means 各分量均对易)

$$\hat{J} = \hat{J}_1 + \hat{J}_2$$

$$\begin{cases} \hat{J}_1 : \{ |j_1, m_1\rangle \} \\ \hat{J}_2 : \{ |j_2, m_2\rangle \} \\ \hat{J} : \{ |j, m\rangle \} \end{cases} \Rightarrow |j, m\rangle$$

(1) 无耦合表象: $\{\hat{J}_1^2, \hat{J}_{1z}, \hat{J}_2^2, \hat{J}_{2z}\}$

本征矢: $|j_1, m_1, j_2, m_2\rangle$

$$= |j_1, m_1\rangle \otimes |j_2, m_2\rangle \rightarrow \text{在自旋那里简写: } |S_1, m_{S1}\rangle \otimes |S_2, m_{S2}\rangle$$

$$\Rightarrow |\pm m_{S1}\rangle \otimes |\pm m_{S2}\rangle \Rightarrow |+\rangle_1 \otimes |+\rangle_2 \Rightarrow |++\rangle_1, 2$$

共同本征态: $\begin{cases} \hat{J}_1^2 |j_1, m_1, j_2, m_2\rangle = j_1(j_1+1)\hbar^2 |j_1, m_1, j_2, m_2\rangle \\ \hat{J}_{1z} = m_1\hbar \\ \hat{J}_2^2 = j_2(j_2+1)\hbar^2 \\ \hat{J}_{2z} = m_2\hbar \end{cases}$

若 j_1, j_2 给定: 维数 $(2j_1+1)(2j_2+1)$

子空间完备性: $\sum_{m_1, m_2} |j_1, m_1, j_2, m_2\rangle \langle j_1, m_1, j_2, m_2| = 1$

(2) 耦合表象: $\{\hat{J}^2, \hat{J}_z, \hat{J}_1^2, \hat{J}_2^2\}$

基矢: $|j, j_z, j, m\rangle$

共同本征态: $\begin{cases} \hat{J}^2 |j, j_z, j, m\rangle = j(j+1)\hbar^2 |j, j_z, j, m\rangle \\ \hat{J}_z = j_z\hbar \\ \hat{J}_1^2 = j_1(j_1+1)\hbar^2 \\ \hat{J}_2^2 = j_2(j_2+1)\hbar^2 \\ \hat{J}_z = m\hbar \end{cases}$

因为描述同一个物理系统

介

若 j_1, j_2 给定, 维数应与无耦合表象一致: $(2j_1+1)(2j_2+1)$

子空间完备性: $\sum_{j, j_z, m} |j, j_z, j, m\rangle \langle j, j_z, j, m| = 1$

$$\hat{J} = \hat{J}_1 + \hat{J}_2$$

★ 当 j_1, j_2 确定时: $j_{\max} = j_1 + j_2$. $j_{\min}$ 受维数限制不一定为 0.

$$\hat{J}_z = \hat{J}_{1z} + \hat{J}_{2z}$$

$$\therefore \sum_{j_{\min}}^{j_{\max}} (2j+1) = (2j_1+1)(2j_2+1)$$

$$\frac{1}{2} (2j_{\max}+1 + 2j_{\min}+1) (j_{\max} - j_{\min} + 1) = (2j_1+1)(2j_2+1)$$

$$\Rightarrow j_{\min} = |j_1 - j_2|$$

$$\begin{matrix} \downarrow & \downarrow & \downarrow \\ m & m_1 & m_2 \\ \downarrow & \downarrow & \downarrow \\ j_1+j_2=j & j_1 & j_2 \end{matrix}$$

j 取值: $j_1+j_2, j_1+j_2-1, \dots, |j_1-j_2|$

m 取值: $j, j-1, \dots, -j$

(二) Clebsch-Gordan 系数:

$$|j_1, j_2, j, m\rangle \subseteq |j_1, m_1, j_2, m_2\rangle$$

$$\therefore |j_1, j_2, j, m\rangle = \sum_{m_1, m_2} \langle j_1, j_2, m_1, m_2 | j_1, m_1, j_2, m_2 \rangle^* |j_1, m_1, j_2, m_2\rangle$$

$$= \sum_{m_1, m_2} \langle j_1, m_1, j_2, m_2 | j, j_z, j, m \rangle |j_1, m_1, j_2, m_2\rangle$$

$$\therefore \langle j_1, m_1, j_2, m_2 | j, j_z, j, m \rangle = C_{m_1, m_2, j, m}^{j_1, j_2} \rightarrow \text{C-G 系数.}$$

↓

$$|j_1, j_2, j, m\rangle = \sum_{m_1, m_2} C_{m_1, m_2, j, m}^{j_1, j_2} |j_1, m_1, j_2, m_2\rangle$$

(三). 两个电子的自旋角动量耦合

$$\hat{S}_1, \hat{S}_2$$

$$\hat{S} = \hat{S}_1 + \hat{S}_2$$

$$\hat{S}_1, \hat{S}_1^z, \hat{S}_2 \Rightarrow | \pm m_s \rangle \Rightarrow | \frac{1}{2} \pm \frac{1}{2} \rangle \Rightarrow \begin{cases} | + \rangle \\ | - \rangle \end{cases}$$

(1). 无耦合表象: $\{ \hat{S}_1^z, \hat{S}_2^z, \hat{S}_1^z, \hat{S}_2^z \}$

$$\Rightarrow | \frac{1}{2} m_s \frac{1}{2} m_s \rangle$$

更简洁的表示:

$$| \frac{1}{2} \pm \frac{1}{2} \frac{1}{2} \pm \frac{1}{2} \rangle \left\{ \begin{array}{l} | + + \rangle \\ | + - \rangle \\ | - + \rangle \\ | - - \rangle \end{array} \right.$$

(2). 耦合表象: $\{ \hat{S}_1^z, \hat{S}_2^z, \hat{S}^z, \hat{S}^2 \} \Rightarrow | \frac{1}{2} \pm \frac{1}{2} S m_s \rangle \Rightarrow | S m_s \rangle$

$$S = S_1 + S_2, S_1 + S_2 - 1, \dots, | S_1 - S_2 |$$

$$= 1, 0$$

$$\Rightarrow \begin{cases} S=1, m_s = 1, 0, -1 & \text{自旋三重态} \\ S=0, m_s = 0 & \text{自旋单态} \end{cases}$$



$$| S m_s \rangle: S=1, | 1 1 \rangle = | + + \rangle = | + \rangle_1 | + \rangle_2 \quad (\text{实际上是直积})$$

$$| 1 -1 \rangle = | - - \rangle = | - \rangle_1 | - \rangle_2$$

$$| 1 0 \rangle = \frac{1}{\sqrt{2}} (| + - \rangle + | - + \rangle)$$

$$S=0, | 0 0 \rangle = \frac{1}{\sqrt{2}} (| + - \rangle - | - + \rangle)$$

四. 带电粒子在电磁场中运动的方程.

描述电磁场:

$$1. \begin{cases} E \\ B \end{cases}$$

$$2. \begin{cases} \vec{B} = \vec{\nabla} \times \vec{A} \\ \vec{E} = -\vec{\nabla} \varphi + \frac{\partial \vec{A}}{\partial t} \end{cases}$$

$$\text{用 } \varphi, \vec{A} \text{ 描述: 有规范变换 } \begin{cases} \vec{A}' = \vec{A} + \vec{\nabla} f \\ \varphi' = \varphi - \frac{\partial f}{\partial t} \end{cases}$$

经典中完整守恒系: $H = T + V$

$$= \frac{1}{2} \mu v^2 + V$$

$$= \frac{1}{2} \mu v^2 + q\varphi$$

定义: 带电粒子的正则动量: $\vec{p} = m\vec{v} + \underbrace{q\vec{A}}_{\text{电磁场的动量}}$

$$\therefore H = \frac{1}{2\mu} (\vec{p} - q\vec{A})^2 + q\varphi$$

↓

对应量子力学: $\hat{H} = \frac{1}{2\mu} (\hat{\vec{p}} - q\vec{A})^2 + q\varphi \rightarrow \text{但进入微观后还须考虑自旋}$

(一). 无自旋粒子.

$$\hat{H} = \frac{1}{2\mu} (\hat{\mathbf{p}} - q\mathbf{A})^2 + q\varphi$$
$$= \frac{1}{2\mu} \hat{\mathbf{p}}^2 + \frac{1}{2\mu} q^2 \mathbf{A}^2 - \frac{q}{\mu} (\mathbf{A} \cdot \hat{\mathbf{p}} + \hat{\mathbf{p}} \cdot \mathbf{A}) + q\varphi \quad \star$$

而 $A_x = A_x(x, y, z)$.

$$\therefore [A_x, p_x] = i\hbar \frac{\partial A_x}{\partial x}$$

$$\Rightarrow \mathbf{A} \cdot \hat{\mathbf{p}} - \hat{\mathbf{p}} \cdot \mathbf{A} = i\hbar \nabla \cdot \mathbf{A}$$

由横场条件(规范规范): $\nabla \cdot \mathbf{A} = 0$

$$\therefore \mathbf{A} \cdot \hat{\mathbf{p}} - \hat{\mathbf{p}} \cdot \mathbf{A} = 0$$

$$\therefore \hat{H} = \frac{1}{2\mu} \hat{\mathbf{p}}^2 + \frac{1}{2\mu} q^2 \mathbf{A}^2 - \frac{q}{\mu} \mathbf{A} \cdot \hat{\mathbf{p}} + q\varphi$$

代入薛定谔方程可解.

(二). 自旋 $\frac{1}{2}$ 粒子 (电子, 质子...)

$$\text{磁矩: } \hat{\mathbf{M}} = \frac{q}{2\mu} \hat{\mathbf{S}}$$

$$\therefore \text{相互作用能: } U_i = -\hat{\mathbf{M}} \cdot \mathbf{B}$$
$$= -\frac{q}{2\mu} \hat{\mathbf{S}} \cdot \mathbf{B}$$
$$= -\frac{q\hbar}{2\mu} \hat{\mathbf{S}} \cdot \mathbf{B}$$

$$\star \text{ 自旋 } \frac{1}{2} \text{ 的粒子的哈密顿量: } \hat{H} = \frac{\hat{\mathbf{p}}^2}{2\mu} + \frac{1}{2\mu} q^2 \mathbf{A}^2 - \frac{q}{\mu} \mathbf{A} \cdot \hat{\mathbf{p}} + q\varphi - \frac{q\hbar}{2\mu} \hat{\mathbf{S}} \cdot \mathbf{B}$$

$$\text{不考虑轨道运动: } \hat{H} = -\frac{q}{2\mu} \hat{\mathbf{S}} \cdot \mathbf{B}$$

*. 有泡利方程和转动矩阵, 估计不考, 故不写.

(三). 外磁场中的类氢原子.

$$\varphi = \frac{Ze}{4\pi\epsilon_0 r}, \text{ 对电子: } q = -e.$$

$$\therefore \text{电子的哈密顿量: } \hat{H} = \frac{\hat{\mathbf{p}}^2}{2\mu} + \frac{1}{2\mu} e^2 \mathbf{A}^2 + \frac{e}{\mu} \mathbf{A} \cdot \hat{\mathbf{p}} - e\varphi + \frac{e\hbar}{2\mu} \hat{\mathbf{S}} \cdot \mathbf{B}$$

考虑轨道角动量和自旋角动量的相互作用.

$$\text{引入 } \begin{cases} \text{托马斯修正: } U_t = \frac{1}{2} \nabla^2 \varphi \hat{\mathbf{L}} \cdot \hat{\mathbf{S}} & \rightarrow \text{只有外磁场, 原子内部磁场均忽略} \\ \text{相对论修正: } & \rightarrow U_t \text{ 比 } c \text{ 小 } 2 \text{ 个量级, 忽略.} \end{cases}$$

$$\therefore \hat{H} = \frac{\hat{\mathbf{p}}^2}{2\mu} + \frac{1}{2\mu} e^2 \mathbf{A}^2 + \frac{e}{\mu} \mathbf{A} \cdot \hat{\mathbf{p}} - e\varphi + \frac{e\hbar}{2\mu} \hat{\mathbf{S}} \cdot \mathbf{B} + U_t$$

$$= H_0 + \frac{1}{2\mu} e^2 \mathbf{A}^2 + \frac{e}{\mu} \mathbf{A} \cdot \hat{\mathbf{p}} + \frac{e\hbar}{2\mu} \hat{\mathbf{S}} \cdot \mathbf{B} + U_t$$

$$\downarrow$$
$$H_0 = \frac{\hat{\mathbf{p}}^2}{2\mu} - e\varphi \text{ (库仑能)}$$

设 $\vec{B} = B \hat{e}_z$

取 $\vec{A} = \frac{1}{2} \vec{B} \times \vec{r}$

$$\begin{cases} A_x = -\frac{y}{2}B \\ A_y = \frac{x}{2}B \\ A_z = 0 \end{cases}$$

代回 $\hat{H}$ 中有: $\hat{H} = \hat{H}_0 + \frac{e^2 B^2}{8\mu} (x^2 + y^2) + \frac{eB}{2\mu} (-y\hat{p}_x + x\hat{p}_y) + \frac{eB}{\mu} \hat{S}_z + g \hat{L} \cdot \hat{S}$

$\hat{H} = \hat{H}_0 + \frac{e^2 B^2}{8\mu} (x^2 + y^2) + \frac{eB}{2\mu} (\hat{L}_z + 2\hat{S}_z) + g \hat{L} \cdot \hat{S}$
 很小, 忽略 ↓ 电子的自旋磁矩

原子磁矩: $M_z = -\frac{\partial \hat{H}}{\partial B_z}$

$$= -\frac{e^2 B}{4\mu} (x^2 + y^2) - \frac{e}{2\mu} (\hat{L}_z + 2\hat{S}_z)$$

抗磁性 固有磁矩

⇒ 可解释材料的顺、抗磁性

(四). 光谱的精细结构

1. 无外场: $B=0$

$\hat{H} = \hat{H}_0 + g \hat{L} \cdot \hat{S}$

式中: $\hat{L} \cdot \hat{S} = \frac{1}{2} (\hat{J}^2 - \hat{L}^2 - \hat{S}^2)$, $\hat{J} = \hat{L} + \hat{S}$

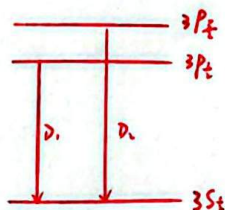
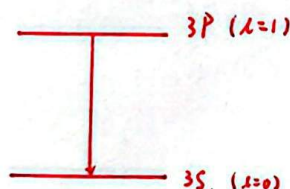
$|4\rangle = |n, l, m\rangle$

$\hat{H}_0$ 作用在 $|4\rangle$ 上: $\frac{1}{2} [j(j+1) - l(l+1) - \frac{3}{4}] \hbar^2$, $j = l \pm \frac{1}{2}$ (不考虑 g)

$\hat{H}_0 \sim E_{nl}$

能级: $E_{nl} \sim E_{nl} + \frac{g}{2} [j(j+1) - l(l+1) - \frac{3}{4}] \hbar^2 \Rightarrow E_{nlj}$ ($j = l \pm \frac{1}{2}$)

eg. Na.



选择定则: $\Delta j = 0, \pm 1$

(2). 有磁场

①. 简单塞曼效应 → 外磁场较强

$\hat{H} = \hat{H}_0 + \frac{eB}{2\mu} (\hat{L}_z + 2\hat{S}_z) \rightarrow \hat{L}_z$ 与 $\hat{S}_z$ 相互独立, 因此不合并成 $\hat{J}_z$

$E_{nl} \Rightarrow \sim E_{nl} + \frac{eB}{2\mu} (m+1) \hbar$

$\Rightarrow E_{nlmms} = E_{nl} + (m \pm 1) \hbar \omega$ $\omega = \frac{eB}{2\mu}$

★

②. 复杂塞曼效应 → 外磁场较弱

$\hat{H} = \hat{H}_0 + \frac{eB}{2\mu} (\hat{L}_z + 2\hat{S}_z) + g \hat{L} \cdot \hat{S}$ (相互不独立, 需耦合)
 $= \hat{H}_0 + \frac{eB}{2\mu} (\hat{J}_z + \hat{S}_z) + \frac{1}{2} g (\hat{J}^2 - \hat{L}^2 - \hat{S}^2)$
↓
 $\hat{L}_z + \hat{S}_z$

$\therefore E_{nl} \Rightarrow \sim E_{nl} + (m_j \pm \frac{1}{2}) \hbar \omega + \frac{1}{2} g [j(j+1) - l(l+1) - \frac{3}{4}] \hbar^2$

$\Rightarrow E_{nlj m_j m_s}$ ★

0. 这里只适用于单电子, 而原子物理中多考虑多电子, 因此对应起来有些不同

第十章. 近似解法.

结论:

1. 本征值的1级修正项: $E_n^{(1)} = H'_{nn}$ (无论 H' 是什么形式, 对角线上的元素即为 $E_n^{(1)}$).

$$2. E_k^{(2)} = \sum_{n \neq k} \frac{|H'_{kn}|^2}{E_k^{(0)} - E_n^{(0)}}$$

$$\therefore E_k \approx E_k^{(0)} + E_k^{(1)} + E_k^{(2)}$$

2. 本征矢的一阶修正项: $\psi_k^{(1)} = \sum_{n \neq k} \frac{H'_{nk}}{E_k^{(0)} - E_n^{(0)}} \psi_n^{(0)}$

$$\therefore \psi_k \approx \psi_k^{(0)} + \psi_k^{(1)}$$

简并态微扰论: $E_n^{(1)} = E_n$, $|\psi_n^{(1)}\rangle = \sum_{\alpha} C_{n\alpha}^{(1)} |\psi_{n\alpha}\rangle$

若要确定 $C_{n\alpha}$, 解: $\sum_{\alpha} H'_{n\beta n\alpha} C_{n\alpha}^{(1)} = \bar{E}_n^{(1)} C_{n\beta}^{(1)}$ 即: $H' \begin{pmatrix} \end{pmatrix} = \bar{E}_n^{(1)} \begin{pmatrix} \end{pmatrix}$

↓
可解出 $\bar{E}_n^{(1)}, |\psi_n^{(1)}\rangle$

$$\therefore |\psi_n^{(1)}\rangle = \sum_{\alpha} C_{n\alpha}^{(1)} |\psi_{n\alpha}\rangle$$

$$\bar{E}_n = \bar{E}_n^{(0)} + \bar{E}_n^{(1)}$$

变分法: 即求: $\delta \bar{H} = \delta \left(\frac{\langle \psi | \hat{H} | \psi \rangle}{\langle \psi | \psi \rangle} \right) = 0$. $\rightarrow$ 解出参数, 代入 $|\psi\rangle$ 中. 例 $\bar{E}_0 = \bar{H} = \langle \psi | \hat{H} | \psi \rangle$ ^{基态}

* 无限深势阱: $E_n = \frac{n^2 \pi^2 \hbar^2}{8\mu a^2}$

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi}{a}x\right)$$

* 动量的本征矢在坐标表象下: $\varphi_p(x) = \frac{1}{\sqrt{2\pi\hbar}} e^{\frac{i}{\hbar} p \cdot x}$

$$\text{-1维: } \varphi_p(x) = \frac{1}{\sqrt{2\pi\hbar}} e^{\frac{i}{\hbar} p x}$$

* δ 函数: $\delta(x-x_0) = \frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} e^{\frac{i}{\hbar} p(x-x_0)} dp$

$$\delta(p-p_0) = \frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} e^{\frac{i}{\hbar} x(p-p_0)} dx$$